

Humanoid Locomotion via EMG Latent Action Priors and Muscle Synergies

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1. Abstract

Human locomotion emerges from complex muscle dynamics and high-dimensional neuromuscular control, making it challenging to replicate using standard reinforcement learning methods. While these methods are capable of discovering a coordinated walking gait, the result is often unnatural and fails to replicate real human movement. We approach this problem by proposing an inductive bias that constrains the action space using muscle synergies from surface electromyography (sEMG) signals. This is recorded from a subject walking in different ambulation modes, rather than from a policy trained on the same simulated body. We test how far this biologically grounded prior, blended with the policy's own learned action, advances natural gait across three musculoskeletal models of increasing complexity. To isolate the prior's effect, our reward specifies only a target center-of-mass (COM) velocity without tuning for natural gaits.

Incorporating such a prior improved several lower-limb kinematic fidelity measures over unconstrained baselines for all models. The exact joint metrics improved and the optimal prior-residual weighting varied by body. The prior did not yield exploration efficiency or task return benefits. For the more complex models, the prior successfully guides hamstring and hip-extensor activation magnitude and timing where unconstrained baselines fail. It is evident that the prior helps provide this structure in cases where the task does not demand this coordination. Remarkably, these benefits persist despite inherent signal noise, embodiment mismatch, and incomplete mapping of muscle actuators in simulation. Our findings demonstrate a promising avenue for incorporating neurophysiological structure from sEMG into reinforcement learning to improve biomechanical fidelity. Videos and code are available at: https://github.com/la2pixel/LEAPS

Keywords: Musculoskeletal Simulation · Muscle Synergies · Deep Reinforcement Learning · Motor Control

2. Introduction

Humans walk with a level of adaptability and robustness that current legged robots do not match (Burden et al., 2024). Much of this comes from the central nervous system (CNS), which coordinates dozens of muscles at every step to keep the body stable on unpredictable ground conditions. Simulating this coordination in musculoskeletal models is useful for multiple fields of research: it underpins predictive simulation for clinical gait analysis, the design of assistive devices, and bio-inspired robot control.

Deep Reinforcement learning (DRL) can now produce stable walking in complex musculoskeletal models, without expert demonstrations (Schumacher et al., 2025). Muscle actuation makes this considerably harder than torque control (see Section 3.1). In particular, the body has more muscles than degrees of freedom, so many different activation patterns produce the same joint motion (Bernstein, 1967). An RL controller is free to pick any of them, so a gait that looks human can still be driven by muscle activity and joint loads that humans do not use.

There are several public biomechanics datasets (Jung, 2025) containing surface electromyography (sEMG) signals alongside kinematics and kinetics from humans walking in different conditions. The sEMG signals contain the human muscle coordination, timing and co-activation structure, and a task like walking can be captured just by a handful of components that explain the most variance. These are called muscle synergies, which result due to motor modules that the CNS recruits to execute a descending command, in this case, to keep walking (Alessandro et al., 2013; Geyer & Herr, 2010).

They have seen little use in musculoskeletal RL, mainly for two reasons. sEMG is noisy, subject-specific and hard to normalise, and it covers only superficial muscles. It also comes from a different body than the simulated one: in our models only between 22% and 67% of the muscles have a matching channel. The closest work, (Park et al., 2026), constrains RL to synergies, but estimates them from the model by inverse simulation rather than from recorded muscle activity.

This thesis asks how far recorded human sEMG can take musculoskeletal RL when used as a prior, despite these limitations. We use recordings of one participant from the Camargo dataset (Camargo et al., 2021) and build on latent action priors (LAP) (Hausdörfer et al., 2024) that change how the action space itself is represented. In our case, the prior is a frozen decoder trained on walking sEMG that maps a low-dimensional latent action to the model's muscles. A learned

residual covers the muscles and corrections to the embodiment the decoder hasn't learned offline. A weight w sets the balance between the two. We train controllers with this prior on a 2D planar saggital model with 18 muscles and two 3D models with 22 and 90 muscles, and compare them with pure MPO (Abdolmaleki et al., 2018) and DEP-MPO (Schumacher et al., 2023), a state of the art controller for muscle-actuated locomotion. A residual-only null prior and an untrained decoder of the same size separate the effect of the sEMG content from that of the decoder architecture.

Our approach didn't yield task performance in RL. It does not improve return or sample efficiency on any model, and constraining the policy more tightly lowers both. Its effect is on how the models walk. With a suitable choice of w, joint kinematics and joint moments move closer to human reference data in both magnitude and timing, on all the three body models. On the 3D models with high muscle redundancies, the prior recovers human-like hamstring and hip abductor timing that the baselines do not.

In summary, our contributions are:

- We propose an action prior for musculoskeletal RL learned from recorded human sEMG, including a mapping from EMG channels to model muscles when the recordings cover only part of the model.
- We show that our approach does not improve return or sample efficiency, but brings joint kinematics and joint moments closer to human reference data across three models of increasing complexity.
- We demonstrate that on the most redundant 3D models, the prior recovers human-like proximal muscle timing that unconstrained baselines fail to reproduce.

We additionally show controls that separate the contribution of the sEMG content from that of the decoder architecture, and an analysis of how the prior weight w trades task performance against human-like coordination.

3. Literature Review

3.1 Learning Locomotion Policies

Before deep reinforcement learning (DRL), gait in musculoskeletal models was generated primarily through hand-designed controllers – most notably reflex-based models, where individual muscles respond directly to local sensory feedback (Geyer & Herr, 2010), as well as trajectory optimization. DRL replaces this control structure with a policy learned through interaction within the simulation (Song et al., 2021). However, compared to torque-actuated robots, musculoskeletal models pose a harder control problem. Muscles can only contract, so each joint must be driven by antagonistic muscle groups whose forces cancel out. Although neural excitation (typically reflected in sEMG) is a good proxy for the force exerted, it can't be measured directly. This is because muscle activation acts as a brief low-pass filter, and muscle force depends nonlinearly on length and contraction velocity (Winter, 2009). Finally, these systems are overactuated, with more muscles than degrees of freedom and many muscles spanning two joints with the same joint motion can be produced by many activation patterns.

3.1.1 Reward Design and Curricula

Natural walking can be achieved even in complex models with upto 90 muscles by meticulously tuning key biologically motivated reward terms (Schumacher et al., 2025) or by combining such rewards with curricula that transition from standing to walking (Weng et al., 2021). While Badie et al. (2025) employed a simpler reward function, they utilized a developmental curriculum inspired by human growth by scaling the body model from child to adult proportions while progressing the task from balance to locomotion.

3.1.2 Structured Exploration in overactuated systems

However, a well-designed reward is only effective if the policy successfully discovers the targeted behavior. In musculoskeletal models, this search is fundamentally difficult because independent action noise largely cancels out across antagonistic muscles which only produces minimal movement. To address this, DEP-RL (Schumacher et al., 2023) implements differential extrinsic plasticity (DEP) which is a rule in neuroscience originally proposed to model the self-organization of sensorimotor skills. Unlike learned exploration strategies, DEP requires no prior

training and generates coordinated movements that drive rich exploration of the state space early in learning. DEP-RL pairs this exploration mechanism with an agent trained via the off-policy Maximum a Posteriori Policy Optimization (MPO) algorithm and achieves sample efficiency and robustness on musculoskeletal tasks compared to prior methods.

Other methods leave the standard RL setup intact but alter the structure of the exploration noise itself. Introducing noise that is correlated across time (Eberhard et al., 2023) or across actuators (Chiappa et al., 2023). This yields more coherent movements than independent noise applied to each muscle per timestep, enabling more effective exploration.

3.1.3 Imitation learning.

Imitation learning guides the policy towards human-like gait with recorded human motion. In DeepMimic (Peng et al., 2018), the policy is rewarded for tracking a motion capture clip, and Lee et al. (2019) later applied the same principle successfully to a full-body model with several hundred muscles. Adversarial methods such as AMP (Peng et al., 2021) replace the hand-designed tracking reward with a discriminator trained to distinguish simulated from recorded motion, so that the policy learns a style rather than a single trajectory. For musculoskeletal models, benchmarks such as LocoMuJoCo (Al-Hafez et al., 2023) already provide motion capture datasets and imitation baselines for a variety of tasks.

Imitation learning approaches do constrain joint kinematics from targeted rewards or adversarial feedback. However, like all the approaches (Sections 3.1.1, 3.1.2) it lets neuromuscular control, internal muscle activation patterns and joint loading are left entirely for the policy to resolve. A controller in simulation can therefore help walk stably and even look natural while coordinating its muscles in ways no human would.

3.2 Muscle Synergies

How the central nervous system selects among a set of valid patterns to execute a given movement is known as the degrees-of-freedom problem (Bernstein, 1967). The most popular hypothesis is that the CNS does not control each muscle independently, but activates groups of muscles together through a handful of shared patterns called muscle synergies. Each synergy fixes the relative activation of a group of muscles, and movement is produced by modulating a few synergy activations over time instead of every muscle separately.

Evidence for this idea comes mostly from decomposing recorded electromyog-

raphy (EMG). In frogs, the muscle activity of natural kicking behaviours can be reconstructed from combinations of a few synergies (d'Avella et al., 2003). Human walking shows the same low dimensionality: the activity of many leg and trunk muscles is explained by five basic activation patterns, each timed to a phase of the gait cycle. Synergies are usually extracted with linear methods such as principal component analysis (PCA) or non-negative matrix factorisation (NMF). Spüler et al. (2016) proposed nonlinear extraction with autoencoders instead, and found it reconstructed the EMG signal in upper limb muscles more accurately than linear methods, particularly at lower dimensions.

Two caveats limit how synergies can be interpreted. First, the synergies reflected in the EMG signal might also come from the mechanics of the rask rather than from neuromuscular control. Second, surface EMG is a **proxy** measure of muscle activation. It records only superficial muscles and is known to picks up crosstalk from proximal muscles. Despite these limits, EMG is typically combined with musculoskeletal models to estimate muscle forces and joint moments during movement, rather than to guide model control.

3.3 Synergy-Based Action Representations

The Synergistic Action Representation (SAR) framework (Berg et al., 2023) first trains a policy on a simpler task like levelground walking, and extracts synergies from the muscle activations it produces in simulation. On the target task, the policy outputs both synergy activations and a full action, which are blended with a fixed weight. This made harder tasks, such as walking on uneven terrain, learnable within a training budget in which standard RL did not succeed. DynSyn (He et al., 2024) derives synergies from the dynamics of the model itself and adapts them to the task during learning, which improved sample efficiency and robustness across several musculoskeletal models.

Park et al. (2026) obtained synergies from one participant whose walking trials were recorded with exact electrodes expected in the 90-muscle model. They used inverse musculoskeletal simulation to estimate the muscle activations that would reproduce this motion in the model and structured the action space of the human body model. Compared to an unconstrained controller, the synergistic constrained controller matched human kinematics, joint moments and muscle timing more closely, although the unconstrained controller reached a higher reward. Their synergies, however, are still computed by the model. Inverse simulation works backwards from motion and resolves muscle redundancy by optimization, so the resulting activations reflect the model and the optimization criterion rather than measured human coordination,

3.4 Priors in Robot Learning

Domain-specific priors are typically used to improve sampl efficiency and task performance of the learning process. Reward terms and curricula highlight which gait features are desirable, while structured exploration encodes assumptions about how an overactuated body should be moved, imitation brings in recorded human motion, and synergies restrict the space of the muscle activations. They differ in where the prior acts and in where its knowledge comes from.

Our approach is inspired by latent action priors (LAP) (Hausdörfer et al., 2024) which was successful at improving sample efficiency, higher rewards and improve performance on transfer tasks for torque actuated robots. They only needed expert actions from a single gait cycle, from hand-designed controllers or trained policies. An autoencoder is trained on the actions of an expert, and its decoder, maps a low-dimensional latent action to the full action space. The decoder is frozen and is employed in the policy to bias the actions towards expert's coordination.

We retain the core structure of LAP but adapt its data source by training the decoder on empirical human walking EMG. Since EMG measurements map to only a subset of the model's degrees of freedom, a remaining residual drives the muscles without a direct channel counterpart. This serves, to the best of our knowledge, as the first action prior for musculoskeletal RL learned from actual surface EMG (sEMG) signals during gait.

4. Methodology

4.1 Methodological Overview

We design a latent action prior as an **inductive bias** that constrains policy exploration toward human muscle coordination patterns. We evaluate how our approach alters learning outcomes compared to unconstrained baselines that discover coordination strategies independently. We hypothesized that constraining the policy with this prior would yield two primary benefits: (1) improvements in learning outcomes, such as faster convergence and higher sample efficiency and (2) enhanced biomechanical fidelity compared to unconstrained baselines.

All analyses compare three groups of variants on all the musculoskeletal models (refer to Section 4.7 for details):

1. Baseline Variants (MPO, DEP-MPO)
2. Prior variants with different residual w variations
3. Control Variants that test whether the coordination structure present in the signal matters, by removing the prior's content or replacing its decoder with an untrained one.

The **methodology** proceeds as follows (summarised in Figure 4.1):

1. Train an autoencoder on sEMG signals from a single subject walking, freezing its decoder to serve as a low-dimensional prior.
2. Map the decoder's EMG channels onto the muscles of each model using anatomical biomechanical principles.
3. Train all variants on each model using Maximum a Posteriori Policy Optimization (MPO). The prior variants operate via the frozen decoder combined with a residual action across all muscles, blended using a weight w.
4. Evaluate all variants using standard reinforcement learning metrics: test return, fall rate, learning speed, and effort.
5. Quantify the biomechanical fidelity of each variant by measuring its alignment with human kinematics, kinetics, and muscle activity patterns.

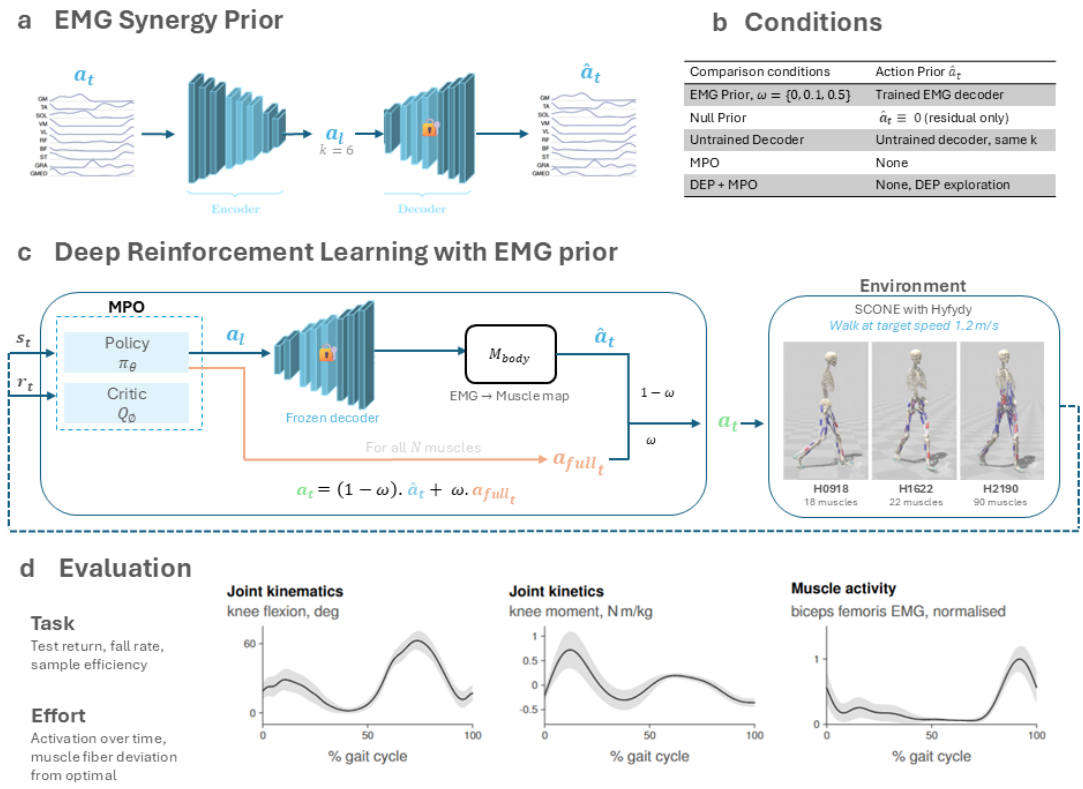


Figure 4.1: **Overview of the EMG prior framework. a.** An autoencoder is trained offline on lower-limb sEMG signals from one person; the decoder is frozen. **b.** Prior, baseline and control variants for comparison. **c.** During training, the policy acts through the frozen decoder for mapped muscles and a residual action, blended with weight w. **d.** Evaluation metrics against human reference from the dataset.

4.2 Preliminaries

Reinforcement Learning. A Markov Decision Process is defined as a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma, \mu_0)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}^+$ is the transition kernel that gives the probability of transitioning from state s to state s' given action a, $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function, γ is the discount factor, and $\mu_0 : \mathcal{S} \to \mathbb{R}^+$ is the initial state distribution. At each step the agent observes a state $s \in \mathcal{S}$, selects an action $a \in \mathcal{A}$ according to the policy $\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^+$, transitions to the next state s' with probability $P(s' \mid s, a)$ and receives the reward $r(s, a)$. We use parametric policies $\pi(a \mid s, \boldsymbol{\theta})$ with parameters $\boldsymbol{\theta}$. Together with the transition kernel and the initial state distribution, the policy defines the stationary state distribution $\mu_\pi(s)$. The probability of a trajectory $\tau_t = \{(s_t, a_t), \ldots, (s_T, a_T)\}$ of length T that starts from s_t and a_t and follows π

thereafter is

$$p_\pi(\tau_t \mid s_t, a_t) = \prod_{t'=t}^{T-1} P(s_{t'+1} \mid s_{t'}, a_{t'})\, \pi(a_{t'+1} \mid s_{t'+1}). \tag{4.1}$$

In muscle-actuated models, the action is the vector of muscle excitations $a \in [0,1]^n$, where n is the number of muscles of the model.

Muscle model. Each muscle is a Hill-type model (Hill, 1938), in which an active contractile element and a passive elastic element act in parallel and together pull on an elastic tendon. The excitation a_i is the neural command to muscle i, but it does not set the force directly. It first drives the activation $\rho_i \in [0,1]$ through first-order dynamics, so that activation lags behind excitation. The force then depends on the state of the muscle,

$$F_i = F_i^{\max}\big[\rho_i\, f_L(\tilde{\ell}_i)\, f_V(\tilde{v}_i) + f_P(\tilde{\ell}_i)\big], \tag{4.2}$$

where $F_i^{\max}$ is the maximum isometric force and f_L, f_V and f_P scale it with the normalized fiber length $\tilde{\ell}_i$ and velocity $\tilde{v}_i$. The same excitation can therefore produce very different forces depending on how long, how fast and how recently activated the muscle is, which is why these quantities are part of the observation (Section 4.6).

Muscle excitation and EMG. Surface EMG measures the electrical activity that the neural command produces in the muscle. The signal id rectified, and a low-pass filtered envelope is applied. This is usually the standard experimental estimate of excitation, used for example to drive EMG-informed musculoskeletal models (Lloyd & Besier, 2003; Buchanan, Lloyd, Manal, & Besier, 2004). This allows a model of EMG envelopes to act as a model of excitations. The correspondence holds for timing and for how the amplitude of each muscle varies over the gait cycle, but not for absolute level. Each channel is normalized separately to the subject's own walking (Section 4.3), not to a maximal contraction, so an envelope value of 1 does not mean a fully excited muscle, and amplitudes are not comparable between muscles. The prior therefore supplies the coordination pattern, and the policy remains responsible for its scale through the residual action (Section 4.7.1).

Our Approach. Here, the policy does not output a directly. It outputs a vector u in a re-parameterized action space $\mathcal{U}$, and a fixed map $f : \mathcal{U} \to \mathcal{A}$ turns u into muscle excitations, $a = f(u)$. Because f has no trainable parameters, it can be treated as part of the environment. The learning problem is then the MDP

$(\mathcal{S}, \mathcal{U}, P_f, r_f, \gamma, \mu_0)$ with $P_f(s' \mid s, u) = P(s' \mid s, f(u))$ and $r_f(s, u) = r(s, f(u))$, and the RL algorithm is used without modification. In the rest of this chapter the policy is written $\pi(u \mid s, \boldsymbol{\theta})$ and the subscript f is dropped. For the baselines $u \in [-1, 1]^n$ and f only clips u to $[0, 1]$. For the latent action prior f is defined in Section 4.7.1.

4.3 EMG Characteristics and Processing Pipeline

4.3.1 Dataset

The prior is learned from the open dataset of Camargo et al. (Camargo et al., 2021), in which $N = 22$ healthy adults (13 male, 9 female; age 21.6 ± 3.6 years, height 1.71 ± 0.07 m, mass 68.5 ± 11.3 kg) walked in four ambulation modes: on a treadmill at speeds from 0.5 m/s in steps of 0.05 m/s, over level ground at self-selected slow, normal and fast speeds, up and down ramps of six inclines between 5.2° and 18°, and up and down stairs of four step heights between 102 and 178 mm. Alongside motion capture, the dataset contains force plate, goniometer, inertial and surface EMG recordings. We use the EMG of a single subject to train the decoder (Section 4.4). The joint angles, moments and ground reaction forces of **six subjects** are the reference for evaluation: AB06, AB08, AB10, AB11, AB23 and AB30, the six male subjects closest to AB06 in height and mass (1.74–1.80 m, 72.6–83.9 kg). Their demographics and stride counts are listed in Appendix 9.3. Muscle activity is compared with all 22 subjects of the dataset.

Electrodes and muscle functions. EMG was recorded from 11 sites on the right leg and trunk: gastrocnemius medialis (GAS), soleus (SOL), tibialis anterior (TA), vastus medialis (VM), vastus lateralis (VL), rectus femoris (RF), biceps femoris (BF), semitendinosus (ST), gracilis (GRA), gluteus medius (GMED) and the external oblique (EO). These are the $m = 11$ channels the decoder reconstructs. They cover the main functional groups of walking: the plantarflexors (GAS, SOL) support the body and push off in late stance, TA lifts the foot in swing and lowers it after heel strike, and the knee extensors (VM, VL, RF) absorb load in early stance. The hamstrings (BF, ST) brake the swinging leg before heel strike, GMED stabilizes the pelvis in the frontal plane during single support, and GRA and EO contribute to hip adduction and trunk stability. Channel counts of this size are typical of gait datasets: surface electrodes only reach muscles close to the skin, neighbouring muscles must be far enough apart to limit crosstalk, and every added sensor lengthens the setup of each participant. Deep muscles such as the iliopsoas and the adductors therefore have no counterpart in the data.

4.3.2 Signal Processing

The raw EMG, sampled at 1000 Hz, is high-pass filtered at 10 Hz, band-pass filtered between 10 and 450 Hz, rectified and low-pass filtered at 6 Hz to obtain the envelope, with zero-phase filters throughout. The envelopes are segmented into strides between consecutive right heel strikes, which the dataset provides from motion capture, and each stride is resampled to 101 points of the gait cycle. Each channel is normalized per subject by the minimum and maximum of the mean treadmill stride at 1.35 m/s, following the dataset's processing scripts, so that each channel lies on a common scale across subjects. A further scaling applied only to the decoder's training data is described in Section 4.4.2. Filter types and orders are listed in Appendix 9.2.

4.3.3 Subject Selection

The decoder is trained on a single subject. Muscle timing differs between people, and a prior pooled over subjects averages these patterns into one that need not correspond to any of them. We select AB06 (male, 20 years, 1.80 m, 74.8 kg), whose body mass is closest to the 74.5 kg of the models. AB06 contributes 1984 strides across all four modes (961 treadmill, 468 ramp, 303 level ground, 252 stair), so the decoder learns activation patterns from a variety of locomotion rather than from one repeated cycle. The strides are split at random into 80% for training and 20% held out. The remaining subjects are not used to train the prior.

4.4 Learning Synergies from EMG

4.4.1 Choice of Synergy Model

Linear synergy models. We write $a \in [0,1]^m$ for one frame of the $m = 11$ EMG channels, since the synergy model is later used to produce muscle excitations; it covers the recorded channels, not yet the n muscles of a model. A synergy model describes each frame as a combination of $k < m$ synergies,

$$a \approx Wh, \tag{4.3}$$

where each column of $W \in \mathbb{R}^{m \times k}$ is a synergy, a fixed weighting of the muscles, and $h \in \mathbb{R}^k$ states how strongly each synergy is active at that instant.

The common methods differ only in the constraints they place on W and a_l. Principal component analysis (PCA) chooses orthogonal synergies that explain as much variance as possible. Independent component analysis (ICA) chooses statistically independent ones. Non-negative matrix factorization (NMF) requires

W and h to be non-negative, so that a synergy can only add activation to its muscles. NMF is the most commonly used method in motor control, which is is also used by Park et al. (2026).

Requirements for an action prior. To serve as an action prior, a synergy model must meet three requirements:

1. Its output must be a valid excitation in $[0, 1]$.
2. A single synergy must be able to represent reciprocal activation, in which one muscle is excited while its antagonist is silent, so that moving along one latent dimension shifts excitation from a muscle to its antagonist. This is the dominant pattern of coordination around a joint in walking.
3. It must reconstruct the EMG well at small k, since that is where it constrains the policy.

No linear method meets the first requirement: PCA and ICA can produce negative outputs, and NMF outputs are non-negative but can exceed one. NMF also fails the second criteria, because with non-negative weights it can produce reciprocal activation only by alternating two synergies in time, never within one.

Autoencoder. We therefore follow Hausdörfer et al. (2024) and extract the synergies with a nonlinear autoencoder (AE). An encoder compresses each frame into a latent action $a_l \in \mathbb{R}^k$, and a decoder D reconstructs it, $\hat{a} = D(a_l)$. Encoder and decoder each have one hidden layer of width $2k$ with tanh activation. The decoder takes the role of W in Eq. (4.3) and the latent action that of h: it maps k synergy activations to m muscles, but nonlinearly, and a sigmoid on its output keeps every reconstruction in $[0, 1]$. The AE is trained with the loss of Hausdörfer et al.,

$$\mathcal{L}(a, \hat{a}, a_l) = \|a - \hat{a}\|_2^2 + \mathcal{L}_{\text{norm}}(a_l), \tag{4.4}$$

$$\mathcal{L}_{\text{norm}}(a_l) = \begin{cases} 0, & \|a_l\|_\infty < \beta, \\ \exp\big((\|a_l\|_\infty/\alpha)^{10}\big) - 1, & \text{otherwise.} \end{cases} \tag{4.5}$$

The first term is the reconstruction error. The second leaves the latent action free while every dimension stays within $\pm\beta$ and penalizes it steeply beyond, with α setting how steeply. This keeps the latent actions in a compact range, which the policy can later cover completely (Section 4.4.3). We differ from the reference approach in two respects: the sigmoid output, since muscle excitations are bounded, and the penalty parameters, $\beta = 0.6$ and $\alpha = 0.9$ instead of their $\beta = 0.8$ and $\alpha = 1.2$, which we tuned for EMG (Section 4.4.2).

	Held-out R^2					Pairs
Method	$k = 2$	$k = 3$	$k = 4$	$k = 6$	$k = 11$	($k = 6$)
PCA	0.47	0.63	0.71	0.87	1.00	3/3
ICA	0.47	0.63	0.71	0.87	1.00	3/3
NMF	0.43	0.60	0.71	0.87	1.00	0/3
AE	0.48	0.68	0.77	0.89	0.97	3/3

Table 4.1: **Comparison of synergy extraction methods** on the pooled cohort (22 subjects, four held out). Pairs: number of the three agonist–antagonist pairs represented by at least one synergy.

Comparison. PCA, ICA, NMF and the AE were compared on the pooled cohort of 22 subjects, with four subjects held out, for $k = 1, \ldots, 11$ (Table 4.1). Reconstruction is measured as the share of variance in the held-out frames that the model explains (R^2). A method is counted as representing an agonist–antagonist pair if at least one synergy weights the two muscle groups with opposite signs. The pairs are TA against GAS and SOL at the ankle, VM, VL and RF against BF and ST at the knee, and GMED against GRA at the hip. Since the AE decoder is nonlinear, it has no single weight matrix; its weights are taken as the change in each muscle when one latent dimension moves away from the centre of the latent space (the Jacobian of D at $a_l = 0$).

We compare the methods at $k = 6$, the dimension used in this work (Section 4.4.2). At this dimension all four reconstruct the held-out EMG equally well ($R^2 = 0.87$), so they differ only in the other two requirements. NMF cannot represent any of the three pairs, since its weights are non-negative. PCA and ICA represent all three, and reconstruct identically, since ICA only rotates the PCA subspace, but their outputs are unbounded. The AE represents all three pairs from $k = 3$ onwards and its output lies in $[0, 1]$ by construction, making it the only method that meets all three requirements.

4.4.2 Decoder Training

The decoder used in all experiments is trained on the strides of AB06 alone, split 80/20 into training and held-out strides (Section 4.3.3). Before training, each channel is divided by its mean per-stride 99th percentile over the training strides and clipped to $[0, 1]$, so that the targets match the range of the sigmoid output. Dividing by a percentile instead of the maximum keeps single-frame artifacts from setting the scale. Each frame is treated independently without temporal context. Our decoder specifies which muscles are active together, while the timing of activation is left to the policy, which chooses a new latent action at every step.

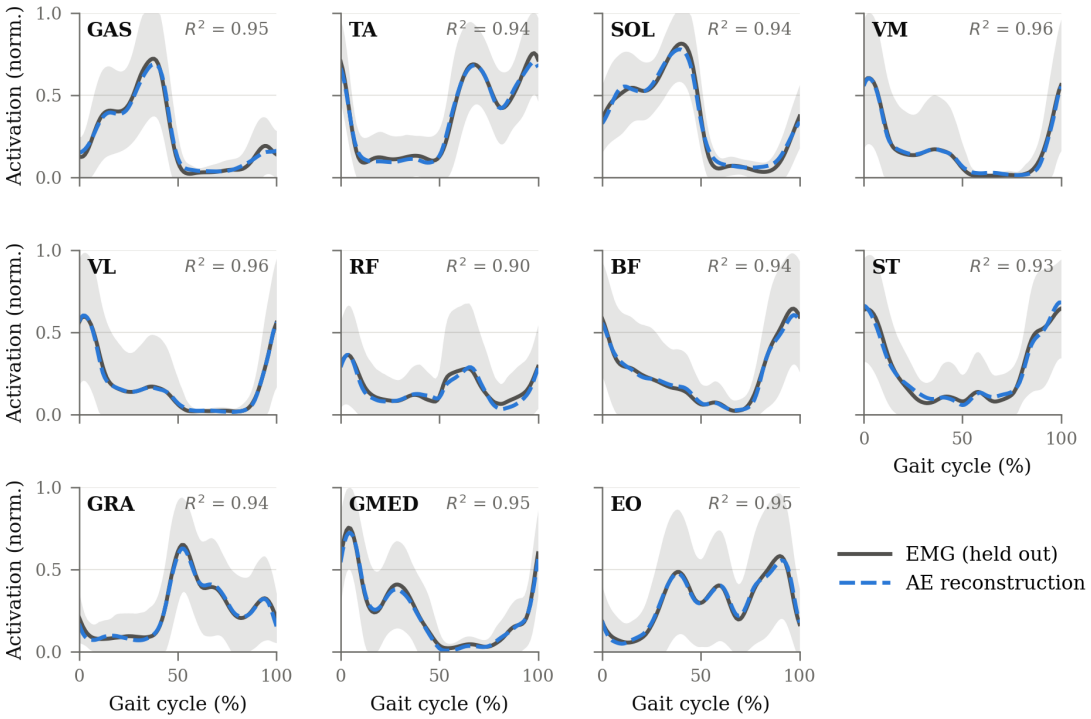


Figure 4.2: **Reconstruction of held-out EMG by the decoder** ($k = 6$, AB06). Mean EMG envelope of held-out strides across all four ambulation modes (solid, shaded ± 1 SD) and the mean AE reconstruction (dashed), over the gait cycle from right heel strike. R^2 per channel is computed over all held-out frames.

k	2	4	6	8	11
Held-out R^2	0.63	0.84	0.94	0.97	0.98

Table 4.2: **Reconstruction of held-out AB06 strides by latent dimension** (mean of three seeds).

Latent penalty. As mentioned above, we tuned our approach from the reference approach originally meant for joint torques of simulated robots. We tuned both for EMG on a grid over the threshold β, the margin $\alpha - \beta$ and the network depth, with three seeds each. Held-out R^2 was flat over a broad region (0.939–0.942), and we use $\beta = 0.6$ and $\alpha = 0.9$ from its centre. A second hidden layer did not improve reconstruction, so the network keeps one hidden layer per side.

Latent dimension. The latent dimension k sets how strongly the prior compresses coordination: a small k imposes more structure on the policy but may lose patterns the muscles need, while a large k approaches the 11 channels it was meant to summarize. On AB06, reconstruction improves quickly up to $k = 6$ and only marginally beyond (Table 4.2). At $k = 6$ the decoder reaches a held-out R^2 of 0.94, 96% of the value at $k = 11$ with about half the dimensions, and we use $k = 6$ for all experiments.

Single subject or pooled. Training on several subjects did not improve the decoder. With the same architecture, the same number of training frames and the same held-out AB06 strides, the single-subject decoder reached $R^2 = 0.94$, against 0.90 when all 22 subjects were pooled, 0.89 when AB06 was pooled with the four subjects closest to it in height and mass, and 0.89 when AB06 was left out. The same ordering holds with AB08 as the target subject. Pooling therefore adds variability between subjects without adding information about the subject the prior is meant to represent. Architecture and training settings are listed in Appendix 9.6.

4.4.3 The Frozen Decoder as Action Prior

After training, the decoder is frozen and the encoder is discarded. During RL, the policy supplies the latent action directly. Its output passes through a tanh, so the latent part of the policy output lies in $u_l \in [-1, 1]^k$. The decoder, however, has only seen latent actions inside the range spanned by the training data, $[a^{\text{lo}}_{l,i}, a^{\text{hi}}_{l,i}]$ for each dimension i, the minimum and maximum over the encoded training frames. For AB06 at $k = 6$, the bounds lie between 0.6 and 0.8 in magnitude in every dimension, between β and α of the penalty. Outside this range the decoder extrapolates, and its output no longer reflects the coordination in the data. The policy output is therefore mapped linearly onto the range,

$$a_{l,i} = \tfrac{1}{2}\left(a^{\text{hi}}_{l,i} + a^{\text{lo}}_{l,i}\right) + \tfrac{1}{2}\left(a^{\text{hi}}_{l,i} - a^{\text{lo}}_{l,i}\right) u_{l,i}, \qquad i = 1, \ldots, k, \tag{4.6}$$

so that every latent action the policy can take lies within the decoder's training domain. The decoded excitations $\hat{a} = D(a_l) \in [0, 1]^m$ are then mapped onto the muscles of the model and blended with a residual action (Section 4.7.1). Each dimension is rescaled independently, so the policy can still reach combinations of latent actions that do not occur in the data, since the latent dimensions are correlated in the training data.

4.5 Policy Optimization (MPO)

We use Maximum a Posteriori Policy Optimisation (MPO) (Abdolmaleki et al., 2018), which has learned robust locomotion efficiently on musculoskeletal models in SCONE in prior work (Schumacher et al., 2023, 2025). All variants use the same learner, so that differences can be attributed to the action space rather than the algorithm; DEP-MPO only adds an exploration scheme (Section 4.7). MPO

extends the classical RL problem with a relative entropy constraint,

$$\max_{q} \; \mathbb{E}_{\mu_\pi(s)}\Big[\mathbb{E}_{q(u|s)}\big[Q^\pi(s,u)\big]\Big] \quad \text{s.t.} \quad \mathbb{E}_{\mu_\pi(s)}\Big[\mathrm{KL}\big(q(u \mid s) \,\|\, \pi(u \mid s, \boldsymbol{\theta}')\big)\Big] \leq \epsilon, \tag{4.7}$$

where $q(u \mid s)$ is a variational distribution, $\boldsymbol{\theta}'$ are the parameters of the target policy, KL is the Kullback–Leibler divergence and ϵ is a hyperparameter. The action-value function of the policy is

$$Q^\pi(s,u) = r(s,u) + \mathbb{E}_{p_\pi(\tau_t | s_t=s,\, u_t=u)}\Big[\sum_{t'=1}^{\infty} \gamma^{t'} r_{t+t'}\Big]. \tag{4.8}$$

It is approximated by a critic $Q(s,u,\boldsymbol{\phi})$, trained to regress the expected-SARSA target

$$y = r + \gamma\, \mathbb{E}_{u' \sim \pi(\cdot|s', \boldsymbol{\theta}')}\big[Q(s', u', \boldsymbol{\phi}')\big], \tag{4.9}$$

where $\boldsymbol{\phi}'$ are the parameters of the target critic and the expectation is estimated from K sampled actions. The solution to Eq. (4.7) is

$$q(u \mid s) \propto \pi(u \mid s, \boldsymbol{\theta}') \exp\left(\frac{Q(s,u,\boldsymbol{\phi})}{\eta}\right), \tag{4.10}$$

which is estimated from the same K samples, with the temperature η obtained by minimizing the convex dual function. Given q, the policy update reduces to supervised learning with the maximum a posteriori objective

$$\max_{\boldsymbol{\theta}} \mathcal{J}(q, \boldsymbol{\theta}) = \max_{\boldsymbol{\theta}} \; \mathbb{E}_{\mu_\pi(s)}\Big[\mathbb{E}_{q(u|s)}\big[\log \pi(u \mid s, \boldsymbol{\theta})\big]\Big] + \log p(\boldsymbol{\theta}), \tag{4.11}$$

where $\log p(\boldsymbol{\theta})$ is a regularizer on the policy parameters. In practice it is a trust region that bounds the KL divergence between $\pi(\cdot \mid s, \boldsymbol{\theta}')$ and $\pi(\cdot \mid s, \boldsymbol{\theta})$, separately for the mean (ϵ_μ) and the covariance (ϵ_Σ) of each action dimension. The target networks follow the online networks by Polyak averaging, $\boldsymbol{\theta}' \leftarrow (1-\tau)\boldsymbol{\theta}' + \tau\boldsymbol{\theta}$ and likewise for $\boldsymbol{\phi}'$.

The replay buffer is the one of Schumacher et al. (2025), which subtracts an adaptive effort cost from sampled rewards once the smoothed episode return exceeds 1000. Under the velocity reward (Section 4.6) the return cannot exceed 1000, so this cost stays zero and the reward is exactly Eq. (4.12). Algorithm 1 summarizes the training loop, and the hyperparameters are listed in Appendix 9.6.

Algorithm 1 MPO for learning a muscle-actuated policy. Blue: steps specific to the latent action prior.

Require: policy $\pi(u \mid s, \boldsymbol{\theta})$, critic $Q(s, u, \boldsymbol{\phi})$, action map f
initialize $\boldsymbol{\theta}$, $\boldsymbol{\phi}$, targets $\boldsymbol{\theta}' \leftarrow \boldsymbol{\theta}$, $\boldsymbol{\phi}' \leftarrow \boldsymbol{\phi}$ and replay buffer $\mathcal{D}$
while the step budget is not reached **do**
  for N_{env} environment steps **do**
    $u_t \sim \pi(\cdot \mid s_t, \boldsymbol{\theta})$
    $a_t \leftarrow f(u_t)$ ▷ muscle excitations
    apply a_t in SCONE, observe r_t and s_{t+1}
    $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s_t, u_t, r_t, s_{t+1})\}$ ▷ store u_t, not a_t
  end for
  for N_{upd} iterations **do**
    sample a batch of B transitions from $\mathcal{D}$
    update $\boldsymbol{\phi}$ towards the target of Eq. (4.9) ▷ critic
    for each state, sample K actions from $\pi(\cdot \mid s, \boldsymbol{\theta}')$
    weight them by Eq. (4.10) ▷ E-step
    update $\boldsymbol{\theta}$ by Eq. (4.11) within bounds ϵ_μ, ϵ_Σ ▷ M-step
    $\boldsymbol{\theta}' \leftarrow (1-\tau)\boldsymbol{\theta}' + \tau\boldsymbol{\theta}$, $\boldsymbol{\phi}' \leftarrow (1-\tau)\boldsymbol{\phi}' + \tau\boldsymbol{\phi}$
  end for
end while

4.6 Exploration with Muscle Actuated Models

Musculoskeletal models. All experiments are run in the SCONE simulator (Geijtenbeek, 2019) with the Hyfydy physics engine (Geijtenbeek, 2021). We use three models that differ mainly in how many muscles act on each degree of freedom (Table 4.3), so that the effect of the prior can be followed as actuation redundancy grows. The name of each model gives its number of degrees of freedom and muscles. **H0918** is planar with nine muscles per leg, close to the minimum needed to actuate hip, knee and ankle in the sagittal plane. **H1622** moves in 3D and adds hip abduction and rotation, which the policy must now also stabilize, with eleven muscles per leg. **H2190** adds the subtalar joint and represents each leg with 45 muscles, several of them split into compartments, so that many muscles with similar function share each joint. In all models, the head, arms and trunk form a single segment without muscles, and the total mass is 74.5 kg.

Each foot is a rigid segment without a toe joint and touches the ground through contact spheres, two per foot in H0918 (heel and toe) and three in H1622 and H2190 (heel, lateral and medial forefoot). Contact follows the Hunt–Crossley model, where a sphere may penetrate the ground slightly and the ground reaction force (GRF) grows with the depth and speed of penetration, with friction opposing sliding.

Model	Motion	DOF	Muscles (per leg)	Observation size
H0918	planar	9	18 (9)	121
H1622	3D	16	22 (11)	161
H2190	3D	21	90 (45)	505

Table 4.3: **Musculoskeletal models in SCONE.** Degrees of freedom (DOF) include the pelvis. In H1622, the lumbar joint is locked. In H2190 it is passive, with a spring and damper and no muscles.

Observations and actions. The policy sets the excitation of every muscle every 0.025 s. The observation consists mostly of quantities a human has access to through proprioception and the vestibular system. For every muscle it contains the fiber length, fiber velocity and force, which correspond to the information carried by muscle spindles and Golgi tendon organs, as well as the excitation and activation. Activation follows excitation with a delay (Section 4.2), so without it the same observation could correspond to different muscle states. The remaining entries are the orientation and angular velocity of the trunk, the position of each foot relative to the pelvis, and the positions and velocities of all degrees of freedom, including the pelvis. The horizontal position of the pelvis is set to zero, so that the policy does not depend on where the model is on the ground. Because the observation grows with the number of muscles, it is more than four times larger for H2190 than for H0918. For every variant, the excitations sent to the simulator are clipped to $[0, 1]$.

Walking task and velocity reward. The task is to walk forward at a target velocity of $v^* = 1.2\,\mathrm{m/s}$, close to the preferred walking speed of adults. Each episode starts from the model's initial pose with Gaussian noise on the joint angles and velocities, so that the policy has to stabilize from a range of states instead of a single one. With probability 0.5 the joint angles and velocities of the left and right leg are swapped, so that episodes start from both the initial pose and its mirror image. An episode ends with a fall when the center of mass (COM) drops below 0.5 m or the trunk drops below 0.9 m, since recovery from these states is unlikely. Otherwise, it ends after 1000 steps (25 s), which covers many gait cycles. The reward depends only on the forward velocity v of the COM,

$$r_t = \begin{cases} \exp\big(-(v_t - v^*)^2\big), & v_t < v^*, \\ 1, & v_t \geq v^*. \end{cases} \tag{4.12}$$

The reward is highest at or above the target, so the maximum return of an episode is 1000. It is intentionally simple and contains no terms for effort, smoothness or ground reaction forces, as such terms favour a particular coordination and would

confound the effect of the prior on the gait. To test whether the effect of the prior depends on this choice, we also train DEP-MPO and the prior with $w = 0.1$ on H0918 and H1622 under the full reward of Schumacher et al. (2025), which encourages natural gait. Its terms and coefficients are listed in Section 5.1.

4.7 Variants for Performance Comparisons

All variants use the same learner (Section 4.5), models, observations and reward. They differ only in what happens between the policy output u_t and the muscle excitations a_t (Figure 4.3, Table 4.4). This isolates three questions. The comparison with MPO asks whether the prior helps over exploring directly in muscle space. The comparison with DEP-MPO asks how it compares to an established method for the same exploration problem. The control variants ask whether any benefit comes from the EMG content of the prior or only from the change of action space.

Variant	Policy output u	Map f	w
MPO	$[-1, 1]^n$	clip to $[0, 1]$	–
DEP-MPO	$[-1, 1]^n$	clip to $[0, 1]$	–
EMG prior	$[-1, 1]^{k+n}$	decoder + residual, Eq. (4.14)	0, 0.1, 0.5
Untrained prior	$[-1, 1]^{k+n}$	untrained decoder + residual	0.5
Null prior	$[-1, 1]^{k+n}$	residual only ($\hat{a} \equiv 0$)	–

Table 4.4: **Compared variants.** The EMG and untrained priors share the map of Eq. (4.14) and differ only in the decoder. The null prior uses the same wrapper, but its residual controls every muscle directly.

4.7.1 Policy with a Latent Action Prior

The policy follows the latent action prior formulation of Hausdörfer et al. (2024). At each step it outputs $u_t = (u_{l,t},\, a_{\text{full},t})$, a latent part $u_{l,t} \in [-1, 1]^k$ and a residual $a_{\text{full},t} \in [-1, 1]^n$ with one entry per muscle. The latent part is rescaled to the latent action $a_{l,t}$ by Eq. (4.6) and decoded by the frozen decoder D into the $m = 11$ EMG channels (Section 4.4.3).

EMG-to-muscle mapping. A fixed matrix $M \in \mathbb{R}^{n \times m}$ assigns each EMG channel to the model muscles it represents, based on anatomical correspondence (Appendix 9.10). A muscle covered by several channels receives their average, and the left-leg muscles copy the rows of the right leg. Muscles without a surface-EMG counterpart receive nothing from the prior; we write $\mathcal{I}$ for the set of muscles that do. The share of covered muscles falls from two thirds of each leg in H0918 (6 of 9) and H1622 (7 of 11) to one quarter in H2190 (11 of 45), so the more detailed the model, the larger the part of it that is controlled without prior content. The prior

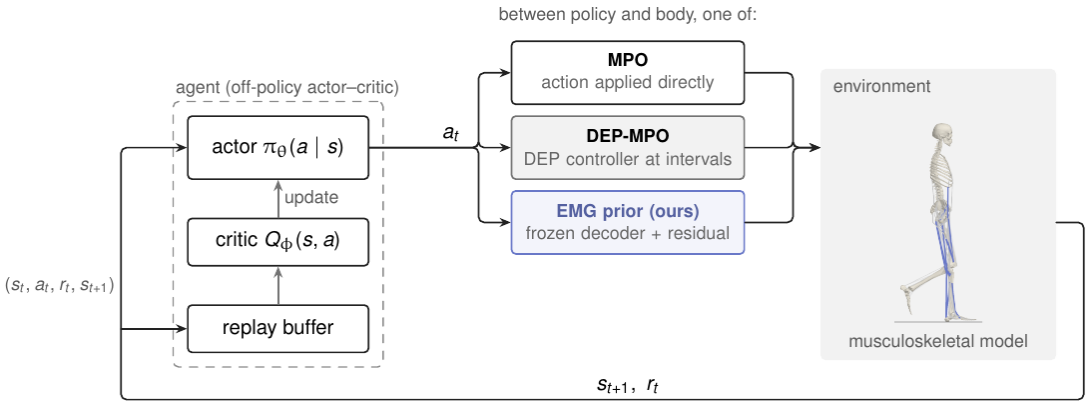


Figure 4.3: **Training loop shared by all variants.** The variants differ only in the map f from the policy output u_t to the muscle excitations a_t.

excitation is

$$\hat{a}_t = M\, D(a_{l,t}) \in [0,1]^n. \tag{4.13}$$

Blending prior and residual. The muscle excitation is a convex combination of prior and residual on the covered muscles,

$$a_{t,i} = \begin{cases} (1-w)\,\hat{a}_{t,i} + w\, a_{\text{full},t,i}, & i \in \mathcal{I}, \\ a_{\text{full},t,i}, & i \notin \mathcal{I}, \end{cases} \tag{4.14}$$

followed by clipping to $[0,1]$. Since the residual lies in $[-1,1]$, it can move a covered muscle up to w above or below the scaled prior. The weight $w \in [0,1]$ therefore sets how far the policy can depart from the prior: at $w = 0$ covered muscles follow the prior exactly, small w keeps them close to it, and at $w = 1$ the prior has no effect. We use $w \in \{0, 0.1, 0.5\}$. Muscles outside $\mathcal{I}$ are always controlled by the residual alone, because the prior has no content for them. Eqs. (4.6), (4.13) and (4.14) form the map f of Section 4.2, which enters Algorithm 1 in line 5.

4.7.2 Baseline Variants

Both baselines act on the muscles directly (Section 4.2). The first is MPO with Gaussian exploration noise on each muscle. The models have several mono- and bi-articular muscles per degree of freedom, and independent noise on each muscle tends to cancel across antagonist pairs, which makes exploration difficult in this over-actuated regime.

The second baseline, DEP-MPO (Schumacher et al., 2023), addresses this problem with differential extrinsic plasticity (DEP), an exploration controller that

couples muscles through the correlations in how their lengths change,

$$\tau_{\mathrm{DEP}}\,\dot{C} = \dot{l}_t\,\dot{l}_{t-\Delta t}^{\mathsf{T}} - C, \tag{4.15}$$

where C is the DEP control matrix, τ_{DEP} sets the averaging window, $\dot{l}$ is the rate of change of the muscle lengths and Δt is a time delay. The output of the DEP controller is scaled by a gain κ. During training, DEP takes over from the policy at random for short intervals, so it changes which actions are explored but not the action space the policy learns in. DEP-MPO is the natural comparison for the prior, since both address the same exploration problem by correlating muscles, but DEP builds the correlations online from the agent's own experience, while the prior takes them from EMG.

4.7.3 Control Variants

The latent action prior changes two things at once. It adds the EMG content of the decoder, and it changes the action space itself: covered muscles are held within w of the prior, and part of the policy output passes through a low-dimensional latent action. Two controls separate these effects.

The **untrained prior** replaces D with a randomly initialized decoder of the same architecture, with the same latent range, mapping and w. Its outputs are still coupled through the shared latent action and constrained in the same way, so it has the structure of the EMG prior but coordination that is not learned from human data. The comparison between the EMG and the untrained prior therefore isolates the EMG content.

The **null prior** sets $\hat{a}_t \equiv 0$ and gives the residual full control of every muscle. It differs from MPO only in the k additional, inactive policy outputs and the wrapper around the environment, and checks that these alone do not change learning.

4.8 Training Protocol

This section describes how the variants are trained and how their task performance is measured, addressing the first hypothesis: whether the prior changes how well and how efficiently the walking task is learned.

4.8.1 Training Runs

Every condition is trained for 2×10^7 environment steps per seed on each model under the velocity reward. Results are reported from training for **three** seeds. The residual weight w is not treated as a hyperparameter to be tuned. It sets how strongly the prior constrains the covered muscles, from fully determined by the

prior ($w = 0$) to half free ($w = 0.5$), and a single value would describe only one point of this trade-off. The EMG prior is therefore trained at $w \in \{0, 0.1, 0.5\}$ on all three models (Section 4.7.3).

Further, MPO, DEP-MPO and the prior with $w = 0.1$ are additionally trained under the full reward scheme on H0918 and H1622 models.(Section 4.6).

Sensitivity runs. To test how the results depend on the design choices of the prior, the EMG prior is additionally trained on H0918 with latent dimensions $k \in \{2, 4, 8, 11\}$ at $w = 0.1$, and with $w \in \{0.2, 0.3\}$ at $k = 6$, with three seeds each.

4.8.2 Evaluation Rollouts

All evaluations use the policy without exploration noise, acting with the mean of its action distribution. During training, the policy is evaluated every $\Delta_{\text{eval}} = 2 \times 10^5$ steps on **20 episodes**. At the selected checkpoint (Section 4.8.3), it is re-evaluated on 100 new episodes for task performance, and a further 8 episodes per seed are recorded with the full simulator state for the gait analyses of Section 4.9. On H2190, where the fall rate is higher, we only report only those episodes lasting atleast 6 seconds.

4.8.3 Checkpoint Selection and Task Performance

The undiscounted episode return $R_s(\ell)$ of seed s at training step ℓ is logged at every training evaluation. On the larger models, the return curves are not monotonous and hence we can't choose the final checkpoint. One checkpoint per seed is therefore selected with a fixed rule, applied identically to every condition.

Checkpoints are saved every 2×10^6 steps, and each checkpoint $\ell \in \mathcal{C} = \{2, 4, \ldots, 20\} \times 10^6$ is scored by the centred moving average of the evaluation return:

$$\bar{R}_s(\ell) = \frac{1}{5} \sum_{\delta=-2}^{2} R_s(\ell + \delta\, \Delta_{\text{eval}}), \tag{4.16}$$

so that a single spike cannot be selected. The checkpoint with the highest score is selected independently for each seed and denoted ℓ_s^*. All policy-level results, including the gait analyses, use the policy at ℓ_s^*.

Task performance is summarized by three quantities.

1. The **return**, which is the mean over the 100 re-evaluation episodes at ℓ_s^* and not the smoothed training curve.

2. The **fall rate**, which is the fraction of these episodes that end in a fall and

3. The **final return**, which is the mean of the last ten training evaluations that show how much performance is lost by the end of the training budget.

When training is stable, as on H0918, the return and the final return coincide and the rule reduces to reporting the converged return; it only changes the result where training is unstable.

4.9 Evaluation of Biomechanical Fidelity

A high return only shows that a model reaches the target speed without falling and gives us no intuition about how the model walks. This section therefore compares the gaits with human walking from the **six evaluation subjects** mentioned in Section 4.3.1 , addressing the second hypothesis (panel d of Figure 4.1).

4.9.1 Metrics for Comparison

The gaits are compared with human walking with three different groups of metrics. **Muscle activity** is the activation of each muscle over the gait cycle, compared with the EMG envelope of the corresponding human muscle trajectories of the evaluation subjects. It shows whether the muscles are used at the right phases. **Kinematics** metrics are the joint angles of hip, knee and ankle, which describe how the body moves regardless of the forces involved. In healthy walking, the knee flexes slightly after heel strike to absorb the load and flexes strongly in swing to clear the foot, and the ankle plantarflexes at push-off (Winter, 2009). **Kinetics** refer to the forces that produce this motion: the net joint moments, the combined turning effect of all muscles crossing a joint, and the vertical ground reaction force. In healthy walking the ground reaction force has two peaks of slightly more than body weight, one during loading and one at push-off, separated by a dip in mid-stance, and the largest joint moment is the ankle plantarflexor moment in late stance (Winter, 2009). A gait can match at one level and not at another because the same joint angles can be produced with different muscles and a muscle can be active at the right time but too weakly to produce the human moment.

4.9.2 Human Reference Data

All human references come from the dataset of the prior (Section 4.3.1), processed into gait cycles in the same way. Muscle activity is compared from treadmill walking within $\pm 0.3\,\mathrm{m/s}$ of the target speed. This is a reasonable deviation as the reward doesn't strictly penalize exceeding target speed for the walk task. Kinematics and kinetics are compared during treadmill walking between 1.1 and $1.3\,\mathrm{m/s}$, because joint angles and in particular joint moments depend on body dimensions.

4.9.3 Extraction of Gait Cycles

All gait measures are computed from the episodes recorded at ℓ_s^* (Section 4.8.2). Only episodes without a fall are used, because an episode that ends in a fall contains incomplete and atypical strides. Gait cycles are segmented at right heel strike, detected when the vertical ground reaction force of the right foot rises above 5% of its peak in the episode, and each cycle is resampled to 101 points. This is the same representation as the human data (Section 4.3.2), so simulated and human cycles are directly comparable.

4.9.4 Muscle Activation Timing

This analysis tests whether the simulated muscles are active at the same phases of the gait cycle as human muscles. For each EMG channel, the activation of the corresponding right-leg model muscle is averaged over all gait cycles of a condition and normalized to its peak, and so is the mean EMG envelope of each of the 22 subjects. Agreement is measured by the Pearson correlation r between the simulated and the human curve, averaged over the 22 subjects. The correlation ignores amplitude, which is not comparable between a normalized EMG envelope and a simulated activation (Section 4.2), and compares only the timing and shape of activation.

Unlike joint moments, the timing and shape of activation depend little on body size, so the comparison is not restricted to the six size-matched subjects of the kinematic reference. Using all 22 subjects instead gives a more robust estimate of human variability, from 231 pairs of subjects instead of 15.. Since humans also differ from each other, and by different amounts for different muscles, the correlation of each simulated muscle is judged against the correlations between these pairs of subjects (Park et al., 2026). A simulated muscle is considered within the human range if its correlation reaches the 5th percentile of the 231 human pairs; a muscle whose timing varies strongly between people is thus held to a lower bar than one whose timing is consistent. However, while testing the comparison to the six reference subjects, we found that it leaves the correlations and the ordering of the variants essentially unchanged.

A match with human EMG is not returned by construction. The decoder is trained on single frames and only fixes which muscles are active together; when they are active is chosen by the policy through the latent action (Section 4.4.2).

4.9.5 Joint Kinematics and Kinetics

Hip, knee and ankle angles, net joint moments normalized by body mass, and the vertical ground reaction force normalized by body weight are compared with the

six reference subjects using two complementary measures. The first is the Pearson correlation r between the mean simulated cycle and the mean cycle of each subject, averaged over the six subjects, as for muscle activity. It compares the shape and timing of a trajectory, independent of its amplitude and offset. The correlation between pairs of reference subjects serves as the human reference value.

The second measure is called the **experimental match**, which is the fraction of the gait cycle in which the mean simulated trajectory $\bar{y}$ lies within one standard deviation of the human mean,

$$E_y = \frac{1}{101} \sum_{g=1}^{101} \mathbb{I}\big[\, |\bar{y}(g) - \mu_y(g)| \leq \sigma_y(g) \,\big], \tag{4.17}$$

where $\mu_y(g)$ and $\sigma_y(g)$ are the mean and standard deviation over the strides of the reference subjects at gait phase g. Unlike the correlation, E_y also penalizes differences in amplitude, and it scales with human variability at every point of the cycle. Because the human joint angles come from goniometers with a subject-specific offset, the mean difference between simulated and human trajectory is removed before E_y is computed, for all quantities alike, so that E_y compares shape, timing and amplitude but not the absolute level.

4.10 Reporting and Scope of Inference

Aggregation. Task performance is computed for every seed and summarized as mean ± SD across seeds, with the individual seeds shown. Learning curves are shown as mean ± SD over seeds, never as the best seed. The gait analyses use the recorded episodes of the all seeds of each condition (5–10 episodes per seed), pooled into one mean cycle per condition. It is to be noted that on H2190, no method sustains a full episode reliably, so task performance is reported as the return at ℓ_s^* together with the fall rate, as there was no convergence observed.

Scope of Inference. Every result is reported per model and per w, never as a single number pooled over models or over values of w, since whether the prior changes how a model walks may depend on how strongly it is imposed. With three seeds per condition, no significance tests are reported. Differences between conditions are interpreted only when they are consistent across seeds and, for kinematics and kinetics, when both measures agree.

5. Results

The results are organized around the two hypotheses of this thesis. We first examine whether the prior changes how quickly and how well the walking task is learned (Section 5.1), and how these results depend on the latent dimension and the residual weight (Section 5.2). Later, we compare the gaits with human walking at three levels: joint kinematics (Section 5.3), joint kinetics (Section 5.4) and muscle activation timing (Section 5.5) along with other key gait parameters.

5.1 Test Performance

H0918. All variants converge to a walking gait and the performance for this model (Table 5.1) only differs in learning speed of the variants. (Figure 5.1, mean $\pm$ SD over two or three seeds). The baseline variants learn fastest. DEP-MPO reaches a return of 990 after 1.6×10^6 steps and MPO after 1.8×10^6, consistent with the faster discovery of coordinated movement that the DEP approach is known for. The null prior learns as fast as MPO (1.8×10^6). Like MPO, its residual controls every muscle directly, so the policy can shape each excitation towards the reward from the first update, and the extra latent outputs and the wrapper do not slow learning.

The prior variants are slower, and **slower the more it is constrained**. It reaches 990 after 2.2×10^6 steps at $w = 0.5$, and after 2.4 and 2.6×10^6 at $w = 0$ and $w = 0.1$, where two thirds of each leg follow the decoder. The untrained prior shows how much the policy depends on the decoder at small w. At $w = 0.1$ it does not learn at all, which is shown by the return which stays at 65 ± 5 for the whole 2×10^7 steps, with a very high fall rate. Since this pattern is observed across models, this variant isn't reported. At $w = 0.5$ the residual can move each covered muscle by up to half the excitation range, and the untrained prior learns as fast as the baselines (1.8×10^6 steps, 999.3 at the selected checkpoint).[1]

H1622. The performance ordering remains consistent with previous observations, though with a wider gap between methods. MPO, DEP-MPO, and the null prior reach a return of 900 within 3×10^6 steps, followed shortly after by the EMG prior at $w = 0.5$. In contrast, the EMG prior requires significantly more training time at lower weights, needing 4.4×10^6 steps at $w = 0$ and 7.0×10^6 steps at $w = 0.1$. The

[1] The untrained decoder was initialized independently in each parallel training environment (Section 4.7.3).

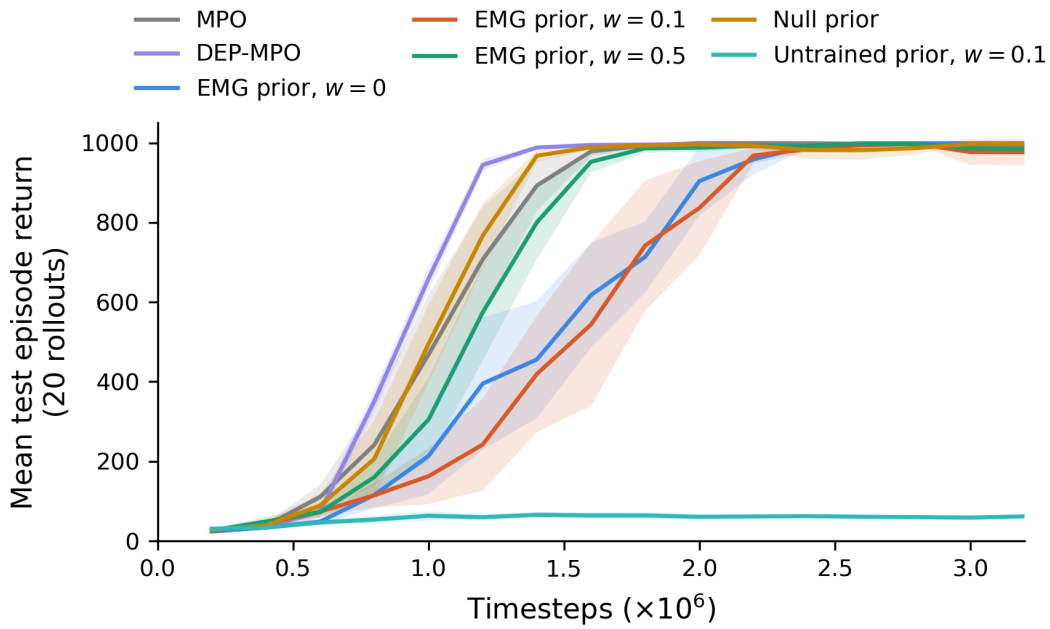


Figure 5.1: **Evaluation return during training on H0918** (mean ± SD over seeds, 20 episodes every 2×10^5 steps, moving average over three evaluations). All variants were trained for 2×10^7 steps; the first 3.2×10^6 are shown, after which every variant stays at the maximum.

same trend pattern as in H0918 is observed for the prior with a $w = 0.1$ residual. At $w = 0.5$, performance closely matches the H0918 baselines, achieving a return of 986.

At $w = 0$, the model shifts from a natural walking gait to a degenerate hopping behavior. Both feet leave and strike the ground simultaneously, keeping the model airborne for roughly 60% of the cycle with virtually no single-support phase. While hopping successfully reaches the target speed, maintaining a high return of 957 ± 3 at $w = 0$ is noticeably less stable. Hence, this variant is not reported for this body model in further analyses.

H2190. No variant converges. Returns at the selected checkpoint lie between 340 and 520, and 50–70% of episodes end in a fall. The baselines reach the highest returns (MPO 505 ± 28, DEP-MPO 519 ± 83). Among the prior variants the order by w is the same as the previous models. In this model, the decoder covers only a quarter of each leg. Hence, the prior constrains the policy significantly less than in H0918 and H1622; however, it still fails to improve performance. The primary challenge of H2190 lies in coordinating the large number of unconstrained muscles per leg, a task for which the prior doesn't help reconstruct. Neither the baseline nor the control variants provide any distinct advantage here, as the sheer

	H0918		H1622		H2190	
Variant	Return	Fall	Return	Fall	Return	Fall
MPO	999.4 ± 0.2	0%	995.2 ± 5.0	0.3%	505.1 ± 27.7	54.5%
DEP-MPO	999.4 ± 0.1	0%	995.2 ± 4.0	0.3%	518.8 ± 82.6	53.0%
EMG prior, $w = 0$	999.8 ± 0.1	0%	957.0 ± 3.3	3.5%	402.4 ± 31.3	58.5%
EMG prior, $w = 0.1$	999.8 ± 0.1	0%	902.2 ± 80.9	9.2%	391.7 ± 40.3	61.2%
EMG prior, $w = 0.5$	999.8 ± 0.0	0%	986.1 ± 2.4	2.0%	440.2 ± 68.8	70.5%
Null prior	999.5 ± 0.2	0%	967.2 ± 17.3	4.3%	350.9 ± 19.2	70.0%

Table 5.1: **Return and fall rate at the selected checkpoint** (mean ± SD, 100 episodes per seed, velocity reward, maximum return 1000). Highlighted values indicate the best metric per column within each model group.

complexity of the model necessitates a more sophisticated reward scheme that isn't tested for this model in the scope of this thesis. Nevertheless, evaluating this setup provides valuable insights into the specific conditions under which the prior is actually beneficial.

Reward for Natural Walking. The velocity reward was chosen because it contains no term that favours a particular coordination (Section 4.6). A prior could therefore look helpful or harmful only because the reward leaves coordination open. To test whether the results depend on this choice, MPO and the EMG prior at $w = 0.1$ were also trained on H0918 and H1622 under:

$$r_t = 10\, r_t^{\text{vel}} - 1.58\, p_t^{\text{mus}} - 0.17\, p_t^{\text{grf}} - 0.13\, p_t^{\text{lim}} - 0.10\, p_t^{\text{smooth}}, \tag{5.1}$$

where r_t^{vel} is the velocity reward of Eq. (4.12), p_t^{mus} is the fraction of muscles with activation above 0.15, $p_t^{\text{grf}} = \max(0,\, F_t - 1.2)$ penalizes a total ground reaction force F_t above 1.2 body weights, p_t^{lim} is the mean absolute torque applied by the joint limits, and $p_t^{\text{smooth}} = \frac{1}{n}\|a_t - a_{t-1}\|^2$ is the mean squared change in excitation between steps. By penalizing excessive muscle recruitment, hard impacts, joint-limit violations, and sudden shifts in excitation, the model collectively favors a natural gait. The maximum achievable return is 10000.

Hence, we can conclude **additional terms do not close the gap**, and the lower performance of the constrained prior does not depend on the velocity-only reward. Under the full reward, the return of both variants declines after the selected checkpoint at the end of the training. This happens because the replay buffer's adaptive effort cost (Section 4.5) becomes active once the return exceeds 1000, so the policy increasingly trades speed for lower effort.

In summary, the coordination structure provided by the prior does not help improve task performance on any model. It learns more slowly than the baselines, and more slowly the smaller w. **Overall, the prior with $w = 0.5$ performs**

Table 5.2: Comparison of return and fall rate between baseline variants (MPO, DEP-MPO) and the prior ($w = 0.1$) on H0918 and H1622 models. Highlighted values indicate the best metric per column within each model group.

Model	Variant	Return	Fall Rate (%)
H0918	MPO ($w = 0.1$)	9600 ± 10	0.0
	DEP-MPO ($w = 0.1$)	9687 ± 60	0.0
	Prior ($w = 0.1$)	9218 ± 82	0.0
H1622	MPO ($w = 0.1$)	8573 ± 213	1.3
	DEP-MPO ($w = 0.1$)	8591 ± 280	1.8
	Prior ($w = 0.1$)	8109 ± 324	5.7

competitively to the baselines on all models. Across successful runs, variants achieve equivalent task returns, rendering task performance metrics insufficient for distinguishing between them. The primary impact of the prior is reflected in the underlying mechanics of locomotion; specifically in how the models walk. We evaluate this through detailed comparisons with reference human gait measurement in the following sections.

5.2 Sensitivity to Latent Dimensions and Residual Weights

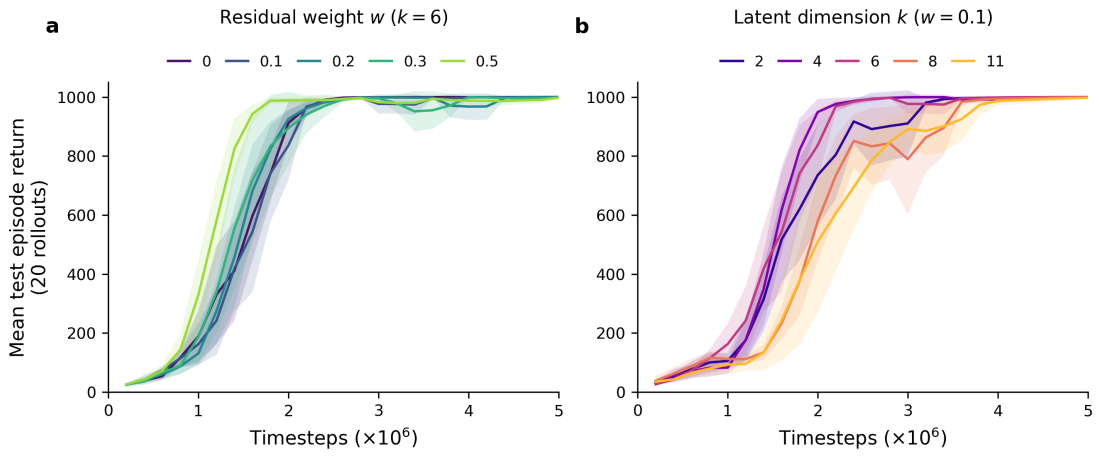


Figure 5.2: **Sensitivity of the EMG prior on H0918** (mean ± SD over three seeds, 20 episodes every 2×10^5 steps, moving average over three evaluations). **a.** Residual weight w with the latent dimension fixed to $k = 6$. **b.** Latent dimension k with the residual weight fixed to $w = 0.1$.

Both the latent dimension k and residual weight w are reported as varied while the other is held fixed. We observed the same trend in the complex models. (Table 5.3).

Residual weight. Learning is stable over the whole range (Figure 5.2a). Every value of w reaches the maximum return, with a final return and the values differ

only in learning speed. From the range of weights tested, we observe that for example, loosening the prior from $w = 0$ to $w = 0.3$ therefore does not speed up learning; only $w = 0.5$ does. This is why our experiments try the three w ranges. Across all three models, $w = 0.5$ learns fastest for all models. This agrees with Hausdörfer et al. (2024), who recommend a higher share of residual $w = 0.5$, for less stable agents such as humanoids.

Latent dimension. All latent dimensions reach the maximum return (final return, but learning speed depends on k non-monotonically (Figure 5.2b). The prior learns fastest at $k = 4$ and $k = 6$; more slowly at $k = 2$ and slowest at $k = 8$ and $k = 11$ when all synergies are used. Here, the dimensions that both reconstruct the EMG well and compress it learn fastest, which supports our choice of $k = 6$ synergies (Section 4.4.2).

Model	w	Steps to threshold	Return
H1622	0.1	4.4×10^6	763 ± 7
	0.5	3.2×10^6	986 ± 2
H2190	0	11.4×10^6	402 ± 31
	0.1	9.4×10^6	342 ± 40
	0.5	7.6×10^6	440 ± 69

Table 5.3: **Sensitivity to the residual weight on H1622 and H2190** ($k = 6$). Steps until the smoothed mean return first reaches 900 (H1622) or 400 (H2190); return and fall rate at the selected checkpoint.

5.3 Simulated Gait Kinematics

Figure 5.3 shows the joint angles and the vertical ground reaction force of H0918 against the six reference subjects, and Table 5.4 reports both measures of agreement for all three models. For easier interpretations, the ground reaction force estimates is reported along with the kinetic measurements.

Overall. The prior variants show substantial improvements in trajectory matching across several metrics. The highest correlations (up to 0.95 at the hip and 0.90 at the knee) approach or align well with human inter-subject correlations. The variants differ most clearly at the ankle, where the baselines consistently struggle (E_y 0.13–0.26 with negative correlations), whereas the EMG prior variants achieve significantly higher match fractions and positive correlations.

H0918. The EMG prior with $w = 0.5$ performs competitively and achieves notable improvements in hip and ankle match fractions compared to both baselines

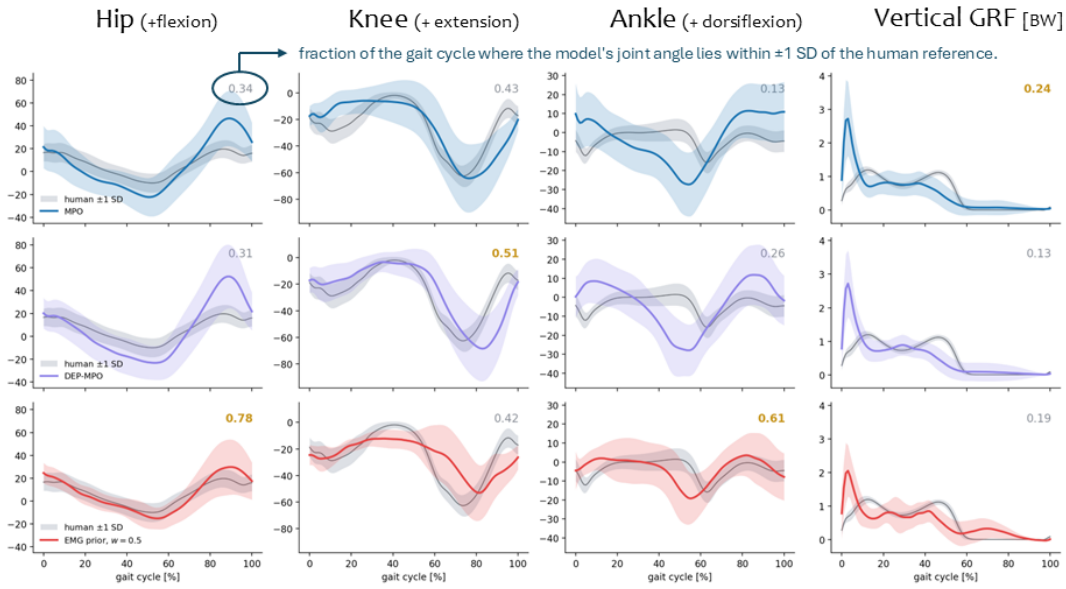


Figure 5.3: **Joint angles and vertical ground reaction forces for key variants.** Reported are the baseline variants and EMG prior at $w = 0.5$ (rows) variants, against the mean ±1 SD of the six reference subjects (grey). Lines: mean over all gait cycles of all seeds, shifted by its mean offset from the human mean as in E_y; shading: ±1 SD over cycles. Values: experimental match E_y, highlighted for the best variant per column. EMG prior at $w = 0.5$ (rows), against the mean ±1 SD of the six reference subjects (grey).

(E_y of 0.78 for the hip and 0.61 for the ankle vs. 0.31–0.34 and 0.13–0.26 for MPO and DEP-MPO). It also maintains strong knee tracking (0.42) and positive ankle correlations (0.32), whereas baseline ankle correlations are negative (−0.06 to −0.12).

H1622. The EMG prior demonstrates clear benefits depending on the weight configuration. At $w = 0.1$, the prior matches the ankle best among all variants (E_y 0.57 against 0.15–0.17 for baselines). At $w = 0.5$, the model achieves strong correlations at both the hip (0.90) and knee (0.90), alongside a solid knee match fraction (0.40).

H2190. The prior continues to show strong ankle tracking advantages at lower weights. At $w = 0.1$, the ankle match fraction peaks at an impressive 0.68 (compared to 0.11–0.21 for baselines). Furthermore, constrained priors ($w = 0$ and $w = 0.1$) sustain positive ankle correlations (0.46 and 0.38) where baseline models fall into negative values, though this advantage disappears at $w = 0.5$ (where the ankle match drops to 0.29 and the correlation drops to −0.37).

		Match fraction E_y			Correlation r		
Model	Variant	Hip	Knee	Ankle	Hip	Knee	Ankle
H0918	MPO	0.34	0.43	0.13	0.89	0.79	−0.12
	DEP-MPO	0.31	0.43	0.26	0.92	0.69	−0.06
	EMG prior, $w = 0$	0.35	0.28	0.25	0.72	0.66	0.20
	EMG prior, $w = 0.1$	0.42	0.25	0.33	0.82	0.82	0.24
	EMG prior, $w = 0.5$	0.78	0.42	0.61	0.90	0.82	0.32
H1622	MPO	0.50	0.29	0.17	0.83	0.86	−0.13
	DEP-MPO	0.27	0.28	0.15	0.75	0.82	−0.17
	EMG prior, $w = 0.1$	0.46	0.28	0.57	0.86	0.65	0.15
	EMG prior, $w = 0.5$	0.45	0.40	0.23	0.90	0.90	0.23
H2190	MPO	0.28	0.39	0.21	0.78	0.85	−0.20
	DEP-MPO	0.31	0.43	0.11	0.86	0.81	−0.19
	EMG prior, $w = 0$	0.36	0.30	0.44	0.95	0.68	0.46
	EMG prior, $w = 0.1$	0.39	0.29	0.68	0.93	0.69	0.38
	EMG prior, $w = 0.5$	0.29	0.42	0.29	0.74	0.76	−0.37
Human (5th percentile)		–	–	–	0.97	0.88	0.74

Table 5.4: **Agreement of the joint angles with human walking.** E_y: fraction of the gait cycle within ±1 SD of the six reference subjects (Eq. (4.17)). r: mean correlation with the six subjects; the human row is the 5th percentile of the correlations between pairs of subjects. H1622 at $w = 0$ is omitted because every seed hops (Section 5.1). Highlighted values indicate the highest metric per column within each model group.

Interpretation. The clearest kinematic effect of the prior is observed at both the ankle and the hip. At the ankle, all baseline models consistently struggle (E_y 0.11–0.26 with negative correlations), whereas the prior variants achieve substantially higher match fractions (up to 0.61 on H0918, 0.57 on H1622 and 0.68 on H2190) alongside positive correlations. This matches where the prior has **direct biological content**. The recorded ankle muscles (TA, SOL and GAS) enforce proper reciprocal activation between dorsi- and plantarflexors to shape the ankle trajectory.

Unlike an assumed absence of effect, the hip also shows sharp improvements – most notably on H0918, where the EMG prior at $w = 0.5$ achieves a peak match fraction of 0.78 and strong correlation (0.90). However, the effectiveness of the prior depends heavily on model morphology and weight configuration. On H2190, the ankle advantage peaks at lower weights ($w \leq 0.1$, reaching $E_y = 0.68$), but this benefit disappears at $w = 0.5$ (where the ankle match drops to 0.29 and correlation falls to −0.37). This indicates that while a strongly weighted prior ($w = 0.5$) works exceptionally well when body composition matches closely (H0918), a higher residual override on mismatched models can disrupt the prior's coordination patterns if pushed too rigidly.

Key Findings

- **Quantitative Improvements:** The proposed prior with a 0.5 residual demonstrated substantial performance gains over baseline methods, particularly showing promising tracking accuracy for hip ($E_y = 0.78$) and ankle ($E_y = 0.61$) angles. Enforcing the prior with a weight of $w = 0.1$ yields performance benefits exclusively in regions with direct muscle actuator coverage.
- **Video Comparisons:** Qualitative comparisons of simulation videos between the baselines and our approach corroborated these findings, providing visual evidence of improved joint trajectory alignments. We also observed simulations with prior variants in general had a **more upright posture**; specifically avoiding the excessive forward torso lean characteristics of the baseline runs.

5.4 Simulated Gait Kinetics

Kinetics describe the forces that produce the motion, which are the net joint moments along with the ground reaction forces. They show whether a joint trajectory produced aligns more with human references, since the same angles can result from different moments. Figure 5.4 shows the joint moments of H0918, and Table 5.5 reports both measures of agreement for all three models.

Overall. The advantage established by the prior variants are not as pronounced in kinetics compared to our observations from kinematics. Hip moments exhibit strong shape correlation across models ($r = 0.77$–0.92), but ankle correlations remain below the human reference range ($r \leq 0.74$, compared to a 5th percentile of 0.93), and knee moment correlations vary widely. Furthermore, every variant lands with a large impact peak in the vertical ground reaction force (1.9–3.5 body weights against 1.2 in humans), with limited shape match ($E_y \leq 0.22$).

H0918. The EMG prior at $w = 0.5$ achieves the hip($E_y = 0.53$), highest knee ($E_y = 0.49$) and ankle ($E_y = 0.34$) match fractions for this model, along with the top ankle correlation (0.73) and a positive knee correlation (0.39). The difference could be visually observed in the ankle plantarflexor moment during late stance (Figure 5.4): the $w = 0.5$ prior produces a distinct push-off peak, whereas baseline variants time their peak moments differently or less effectively.

H1622. Baseline models achieve strong hip and knee kinetic match fractions ($E_y = 0.50$–0.64), outperforming the prior at $w = 0.5$ (0.36–0.39). However, the

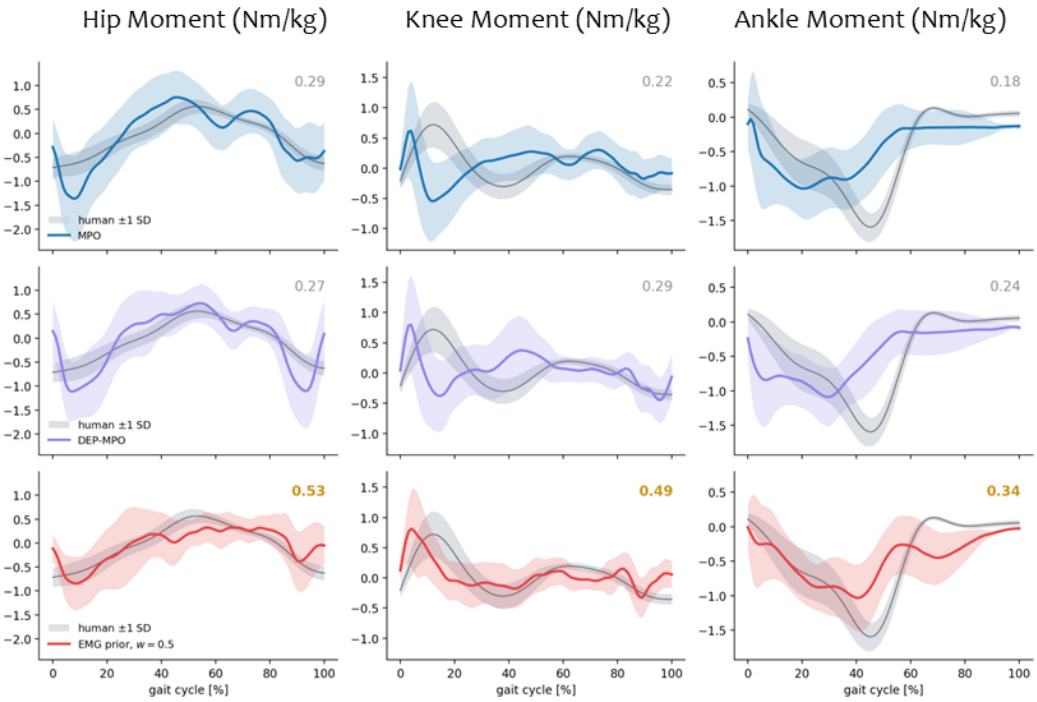


Figure 5.4: **Net joint moments on H0918** for MPO, DEP-MPO and the EMG prior at $w = 0.5$ (rows), normalized by body mass, against the mean ± 1 SD of the six reference subjects (grey). Lines: mean over all gait cycles of all seeds, shifted by its mean offset from the human mean as in E_y; shading: ± 1 SD over cycles. Experimental match E_y is highlighted for the best variant per joint.

prior demonstrates advantages at the ankle. At $w = 0.5$, it yields the highest ankle correlation among all variants (0.74), successfully capturing proper push-off mechanics compared to unguided variants.

H2190. On this model, the prior does not outright improve kinetics. While MPO achieves the highest hip match fraction (0.65), the EMG prior at $w = 0.5$ improves the knee match fraction to 0.60. Ankle correlations for the prior peak at lower weights ($w \leq 0.1$ reaching 0.51), but overall kinetic alignment remains challenging across all variants for this model morphology.

Interpretation. The kinetic measurements mirror the kinematic findings were we observe the most consistent improvements at the ankle. This is again, where the prior contains complete actuator coverage from the dataset. On smaller models like H0918 and H1622, a strongly weighted prior ($w = 0.5$) successfully shapes human-like plantarflexor moments during push-off. However, because these kinetic gains do not uniformly scale with the same weight configurations across larger models like H2190, the exact interaction between prior weighting and kinetic force generation depends strongly on model capacity and morphology.

		Match fraction E_y				Correlation r			
Model	Variant	Hip	Knee	Ankle	GRF	Hip	Knee	Ankle	GRF
H0918	MPO	0.29	0.22	0.18	0.22	0.88	−0.28	0.59	0.59
	DEP-MPO	0.27	0.29	0.24	0.20	0.87	0.02	0.57	0.65
	EMG prior, $w = 0$	0.22	0.23	0.21	0.06	0.89	−0.15	0.59	0.54
	EMG prior, $w = 0.1$	0.24	0.30	0.24	0.15	0.82	−0.15	0.63	0.62
	EMG prior, $w = 0.5$	0.53	0.49	0.34	0.20	0.89	0.39	0.73	0.53
H1622	MPO	0.50	0.64	0.25	0.10	0.91	0.62	0.57	0.62
	DEP-MPO	0.56	0.39	0.18	0.20	0.82	0.55	0.46	0.56
	EMG prior, $w = 0.1$	0.59	0.55	0.19	0.16	0.90	0.32	0.53	0.71
	EMG prior, $w = 0.5$	0.39	0.36	0.31	0.20	0.86	0.51	0.74	0.62
H2190	MPO	0.65	0.57	0.26	0.19	0.89	0.45	0.35	0.50
	DEP-MPO	0.35	0.36	0.21	0.17	0.92	−0.11	0.18	0.56
	EMG prior, $w = 0$	0.16	0.23	0.27	0.14	0.77	−0.31	0.51	0.67
	EMG prior, $w = 0.1$	0.31	0.26	0.24	0.18	0.78	−0.27	0.51	0.71
	EMG prior, $w = 0.5$	0.54	0.60	0.28	0.18	0.90	0.54	0.20	0.61
Human (5th percentile)		–	–	–	–	0.91	0.45	0.93	0.97

Table 5.5: **Agreement of the net joint moments and the vertical ground reaction force with human walking.** E_y: fraction of the gait cycle within ± 1 SD of the six reference subjects (Eq. (4.17)). r: mean correlation with the six subjects; the human row is the 5th percentile of the correlations between pairs of subjects.

5.5 Reconstructing Muscle Activity

Here, we look at whether each simulated muscle is active at the same phases of the gait cycle as the human reference data. We also compare the magnitude. Table 5.6 reports, for each recorded muscle, the correlation r between the simulated activation and the EMG envelope of the 22 subjects, and marks where it reaches within the range of human subject reference bands for each muscle.

Human range interpretations. The plantarflexors and the vasti are active at nearly the same phases in every subject, so their 5th percentile is high (0.67–0.74), and a simulated muscle must match their timing closely to reach it. TA, RF and the hamstrings vary much more between people (0.19–0.28), so a modest correlation is already within the human range. Whether a muscle lies within the range therefore means more for the first group than for the second, and the individual correlations are more informative than the number of muscles within the range.

H0918. On the simpler H0918 model, the prior does not yield a significant benefit in muscle activity. This could be likely because the task is simple enough that the unconstrained baselines already suffice. With $w = 0.5$, the prior times the ankle muscles as well as the baselines (SOL, GAS, and TA lie within the human

range), but the knee extensors are timed worse ($r = 0.14$–0.16 against 0.61–0.64). At $w \leq 0.1$, the ankle muscles are out of phase with human activity (r between -0.38 and 0.00), despite identical task returns. The untrained prior at $w = 0.5$ also times the ankle muscles well, indicating that on H0918, the timing cannot be solely attributed to the EMG content.

H1622. Evaluating muscle activity reconstruction on H1622 reveals clear advantages for our approach where unconstrained baselines compromise. Specifically, when using the prior with a $w = 0.1$ residual, the four highlighted muscles (VM, VL, BF, and ST) perform much better than the baselines, bringing them well within or closer to the human reference bands (reaching $r = 0.86$ for VM, 0.88 for VL, 0.77 for BF, and 0.87 for ST). For instance, examining the reconstruction of the biceps femoris (`BF`) shows that the prior with $w = 0.1$ follows the human pattern much closer than other methods. Conversely, using the prior with a $w = 0.5$ residual only improves the reconstruction of the SOL and GAS muscles, otherwise performing similarly to the baselines. MPO and DEP-MPO baselines perform worse on several components, with MPO's hamstrings active completely out of phase ($r = -0.42$ and -0.41).

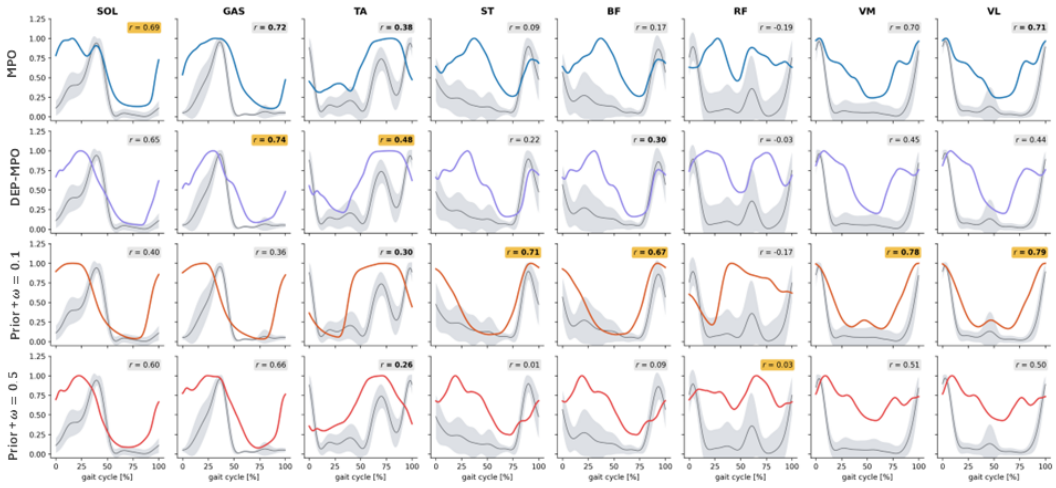


Figure 5.5: **Reconstructions of muscle activation on H1622.** at the selected checkpoint ℓ_s^* for MPO, DEP-MPO and the EMG prior at $w = 0.1$ and $w = 0.5$ (rows), against the mean ± 1 SD EMG envelope of the 22 subjects (grey). Activations are peak-normalized. Lines: mean activation over all gait cycles of all seeds, peak-normalized; Badges: mean correlation r with the 22 subjects; gold, the best variant per muscle; bold, reasonably within the range between human subjects.

H2190. On the most complex and redundant model, the prior at $w = 0.1$ notably improves reconstructs the four highlighted muscles, just like in H1622. Reconstructions of the hamstrings and vasti active at the human phases (e.g.,

BF $r = 0.61$, ST 0.45, VL 0.72), whereas both baselines struggle or yield inverse correlations (e.g., BF $r = -0.36$ to -0.07). At $w = 0.5$, hamstring timing is lost, but both vasti remain within the human range (0.74–0.75), which no baseline reaches. Meanwhile, the baselines perform better on the plantarflexors and RF (e.g., GAS $r = 0.86$ for MPO).

Model	Variant	SOL	GAS	TA	VM	VL	RF	BF	ST
Human (5th percentile)		0.74	0.69	0.22	0.73	0.67	0.19	0.26	0.28
H0918	MPO	**0.78**	0.67	**0.48**	0.62	0.62	−0.13	**0.62**	**0.63**
	DEP-MPO	0.70	**0.72**	**0.65**	0.69	**0.71**	−0.09	**0.48**	**0.42**
	EMG prior, $w = 0$	−0.02	−0.01	−0.33	0.32	0.32	−0.49	**0.38**	**0.31**
	EMG prior, $w = 0.1$	0.26	0.23	−0.22	0.54	0.56	−0.45	**0.29**	0.23
	EMG prior, $w = 0.5$	0.72	**0.75**	**0.53**	0.32	0.32	−0.23	**0.40**	**0.39**
H1622	MPO	**0.79**	0.78	**0.77**	0.72	0.71	−0.13	−0.42	−0.41
	DEP-MPO	0.50	**0.81**	0.66	0.63	0.74	**0.55**	0.08	0.20
	EMG prior, $w = 0.1$	0.52	0.53	0.46	**0.86**	**0.88**	−0.19	**0.77**	**0.87**
	EMG prior, $w = 0.5$	0.74	0.80	0.42	0.48	0.47	0	−0.18	−0.21
H2190	MPO	0.60	**0.86**	**0.56**	0.53	0.54	**0.25**	−0.07	−0.36
	DEP-MPO	0.60	0.51	**0.39**	0.68	0.59	**0.23**	−0.22	−0.16
	EMG prior, $w = 0$	0.72	0.61	0.11	0.00	0.08	−0.74	0.26	−0.02
	EMG prior, $w = 0.1$	0.35	−0.05	−0.22	0.67	**0.72**	−0.66	**0.61**	**0.45**
	EMG prior, $w = 0.5$	0.24	0.22	**0.24**	**0.74**	**0.75**	0.02	−0.04	−0.31

Table 5.6: **Correlations for muscle activation reconstructions for all the models** at the selected checkpoint ℓ_s^*. Mean correlation r with the EMG of the 22 subjects; bold: reasonably within the range between human subjects.

Interpretation. These results indicate that the prior is effective primarily when the task leaves muscle coordination under-determined, leaving multiple degrees of freedom open for muscles that do not strictly need to match human timing to achieve stable locomotion. Because stable walking does not rigidly require human-like electromyography (EMG) patterns across every individual component, unconstrained optimization methods frequently discover alternative, highly functional movement strategies. In models with high structural redundancy (such as H1622 and H2190) featuring numerous actuators per joint, unconstrained reinforcement learning rarely converges independently on reciprocal antagonist timing. The prior addresses this by constraining the policy space to provide appropriate temporal relationships, such as coordinating the vasti and hamstrings. To understand this mechanism more fully, consider why redundancy permits divergence: when a joint is actuated by multiple overlapping muscles, an unconstrained agent can easily find a stable control policy that shifts the workload entirely to one muscle while leaving others dormant or completely out of phase with human behavior. The task reward function only penalizes falling or energy inefficiency, remaining indifferent

to *how* the joint torque is distributed among synergists. The prior bridges this gap by imposing a biological template that forces the model to explore solutions where antagonist muscle pairs fire in correct alternation. Conversely, on simpler models like H0918, where the biomechanics are less redundant and baselines readily converge on human-like timing naturally, the prior provides negligible benefit and can actively degrade coordination performance at lower residual weights ($w \leq 0.1$).

5.6 Other Gait Parameters

The match fractions and scalar correlation values demonstrate whether the whole trajectories are matched to human references with our approach. To see which other properties of the gait differ we look at those gait parameters in detail, that the reward does not constrain.

Speed, cadence, and stance. Because the reward plateaus above the target speed, most variants walk faster than the human reference of 1.2 m/s (ranging from 1.22 to 1.4 m/s). The primary exceptions are the variants using the prior on H2190, which walk slower (0.92–1.05 m/s) in accordance with lower returns. Cadence trends vary across models: on H0918, the baselines step close to the human cadence, while the prior matches this only at $w = 0.5$ (117), stepping much faster and shorter at $w = 0$ (145). Conversely, on the larger models, this pattern reverses. All baselines step substantially faster than humans (130 to 151 steps per minute), whereas the **prior yields closer alignment**, reaching 134 at $w = 0.5$ on H1622 and 108 to 124 across H2190 variants. Regarding gait phases, no variant reproduces the prolonged human stance phase (0.62); all configurations maintain shorter stance durations between 0.47 and 0.56 during locomotion.

Joint load and range of motion. Every variant swings the hip far more than humans do (56–97° against 43°) and loads the knee two to three times more (peak knee moment 0.97–1.45 against 0.45 Nm/kg, while walking). The **prior lowers the knee load** on H0918 (0.77–1.10 against 1.31–1.44) and on H2190 (1.03–1.15 against 1.19–1.45). It also reduces the exaggerated ankle range of motion on H0918 (40–50° against 60–63°; human 28°), but increases the hip range of motion there (88–97° against 79–81°).

Effort. Effort is measured as the mean activation of all muscles,

$$\bar{\rho} = \frac{1}{nT} \sum_{t=1}^{T} \sum_{i=1}^{n} \rho_{i,t}, \tag{5.2}$$

Model	Variant	Speed [m/s]	Cadence [1/min]	Stance	Flight	ROM hip / knee / ankle [deg]	Knee mom. [Nm/kg]	Effort	Fiber dev.
Human		1.20	112	0.62	0	43 / 60 / 28	0.45	–	–
H0918	MPO	1.38	114	0.53	0.01	79 / 62 / 60	1.31	0.51	0.22
	DEP-MPO	1.40	111	0.53	0.01	81 / 71 / 63	1.44	0.49	0.22
	EMG prior, $w = 0$	1.32	145	0.52	0.00	88 / 53 / 41	0.77	0.46	0.24
	EMG prior, $w = 0.1$	1.31	121	0.53	0.00	97 / 70 / 40	0.90	0.46	0.24
	EMG prior, $w = 0.5$	1.33	117	0.52	0.02	95 / 73 / 50	1.10	0.42	0.21
H1622	MPO	1.39	146	0.51	0.02	56 / 72 / 48	1.38	0.46	0.20
	DEP-MPO	1.38	151	0.47	0.08	57 / 74 / 46	1.34	0.44	0.20
	EMG prior, $w = 0$	1.25	154	0.36	0.04	38 / 10 / 35	0.67	0.49	0.21
	EMG prior, $w = 0.1$	1.24	146	0.39	0.02	46 / 23 / 38	0.75	0.46	0.21
	EMG prior, $w = 0.5$	1.23	134	0.52	0.03	67 / 75 / 49	1.29	0.44	0.20
H2190	MPO	1.21	135	0.50	0.05	79 / 87 / 58	1.45	0.47	0.22
	DEP-MPO	1.19	130	0.52	0.05	79 / 83 / 65	1.19	0.47	0.24
	EMG prior, $w = 0$	1.05	108	0.56	0.03	84 / 66 / 62	1.06	0.42	0.23
	EMG prior, $w = 0.1$	1.01	123	0.56	0.01	82 / 62 / 56	1.15	0.44	0.23
	EMG prior, $w = 0.5$	0.92	124	0.51	0.05	59 / 67 / 61	1.03	0.45	0.25
	Null prior	1.08	125	0.53	0.02	73 / 74 / 61	0.97	0.47	0.24

Table 5.7: **Gait parameters.** Stance and flight: fraction of the gait cycle with the right foot on the ground and with neither foot on the ground. ROM: range of motion over the gait cycle. Peak net knee moment measurements are reported.

where $\rho_{i,t}$ is the activation of muscle i at step t, n the number of muscles of both legs, and the sum runs over the middle 90% of each episode, excluding the start and the end; $\bar{\rho}$ is then averaged over episodes. The reward contains no effort term, yet the **prior spends the least muscle activation effort** of all variants on every model: 0.42 on H0918 (at $w = 0.5$, against 0.44–0.52 for the baselines and the null prior), 0.35 on H1622 (at $w = 0.5$, equal to DEP-MPO and below MPO's 0.46) and 0.38–0.41 on H2190 (against 0.47). The null prior, which has the same wrapper but no decoded pattern, spends as much effort as MPO. The saving therefore comes from the decoded pattern, not from the change of the action space.

Muscle Fiber Optimal Range. A muscle produces the most force per unit activation when its fibers are close to their optimal length, at the top of the force–length curve f_L (Section 4.2), and in human walking the major leg muscles are thought to operate near this range. We measure how far the simulated fibers work from it with the normalized fiber length $\tilde{\ell}_i = \ell_i / \ell_i^{\text{opt}}$, where ℓ_i is the fiber length of muscle i and ℓ_i^{opt} its optimal fiber length, through two quantities,

$$D = \frac{1}{nT} \sum_{t=1}^{T} \sum_{i=1}^{n} \left| \tilde{\ell}_{i,t} - 1 \right|, \qquad P = \frac{1}{nT} \sum_{t=1}^{T} \sum_{i=1}^{n} \mathbb{I}\left[0.8 \le \tilde{\ell}_{i,t} \le 1.2 \right], \tag{5.3}$$

the mean deviation from the optimal length and the fraction of time spent within ±20% of it, over all n muscles and the middle 90% of each episode, averaged over episodes.

In all variants, the fibers deviate from their optimal length by 20–25% on

average ($D = 0.20$–0.25) and stay within ±20% of it for about half of the time ($P = 0.44$–0.58). The differences between variants are small and **follow no consistent pattern**. The prior at $w = 0.5$ works slightly closer to the optimum on H0918 ($D = 0.21$ against 0.22) and slightly further from it on H2190 (0.25 against 0.22–0.24).

This insensitivity follows from how fiber length is determined. A muscle–tendon unit's length is set by joint posture and moment arms. Because all variants navigate similar joint ranges, their fibers sweep through comparable lengths. Also, neither the reward nor the prior optimizes for operating length, as the reward lacks an effort or efficiency term and the decoder specifies co-activation rather than operating length. Consequently, the prior does not alter muscle operating lengths; reproducing human-like exploitation of the force–length relationship would likely require an objective that rewards efficiency, such as the effort terms of the full reward.

Summary. The gait parameters confirm and extend the trajectory comparison. The prior does not clearly produce a more natural gait, but it changes **specific properties** in a consistent direction. On every model it spends about 10-15% less effort, and on the smaller and the largest model it helps lower the knee load. On the larger models it brings the cadence closer to human walking.

6. Discussion

6.1 Prior versus Task Performance

The prior was introduced to structure exploration, with the hypothesis that coupling antagonistic muscles according to human co-activation patterns would reduce the cancellation caused by independent exploration in an over-actuated system. Our results do not support this hypothesis. The prior does not accelerate convergence on any model; instead, stronger constraints consistently delay learning (Section 5.1). The null prior, which uses the same wrapper but contains no decoded pattern, learns at the same rate as MPO. This indicates that the slowdown is not caused by the architecture itself. Rather, the policy must first learn to control through the structure imposed by the decoder.

The results suggest that exploration is not the limiting factor. MPO, despite exploring each muscle independently, reaches the target velocity on H0918 and H1622 within a few million steps, while DEP-MPO provides only a marginal improvement. Under the velocity reward, unstructured exploration is therefore already sufficient to discover a gait. For the H2190 model, no variant converges, revealing that the challenge lies in coordinating its highly redundant musculature rather than discovering correlated activations. This is further compounded by the fact that the prior covers only around 22% of each leg.

More importantly, the transferred prior constrains exploration without necessarily pointing toward task-relevant solutions. The decoder is learned from human motion, so it captures plausible human co-activation patterns but not necessarily patterns that are effective for these musculoskeletal models. Latent exploration is consequently restricted to the directions represented by the decoder, within $\pm w$ of the decoded excitation. The policy must learn how to combine these directions with residual corrections before the human-derived structure becomes useful for locomotion. This differs from expert-derived priors such as those of Hausd"orfer et al. (Hausdörfer et al., 2024), where the prior is obtained from behavior that already solves the target task. In that setting, the prior can guide exploration toward useful solutions; here, transferring the prior across bodies instead constrains the search space.

Thus, the prior does not systematically improve task performance, consistent with Park et al. (Park et al., 2026), who found no reward advantage of synergy-based over independent muscle control. Its effect is instead on the coordination

selected once a gait is found.

6.2 Where the Prior Changes the Gait

Although the prior does not consistently improve task performance, it moves specific aspects of the gait towards human walking. These effects are concentrated in muscles represented by the prior and in coordination patterns that are weakly constrained by the velocity objective. Their magnitude and control depend on the residual weight w, with stronger constraints producing clearer changes in lower-limb kinematics and kinetics and overall, the tighter constraints preserving greater flexibility in muscle activation timing.

The 11 EMG channels show reciprocal coordination between major muscle groups including dorsiflexors and plantarflexors and hamstrings and knee extensors (Section 4.4.1). When projected onto the musculoskeletal models, its effects depend on both model redundancy and residual strength:

- **Kinematic and kinetic effects at higher residual share ($w = 0.5$):** On H0918, the stronger prior produces the clearest improvements in trajectory matching, yielding high match fractions at the hip ($E_y = 0.78$) and ankle ($E_y = 0.61$). It is also the only variant to produce a plantarflexor push-off moment with human-like magnitude and timing during late stance (Sections 5.3 and 5.4). Qualitatively, these variants maintain a more upright trunk posture than the unconstrained baselines, which exhibit greater forward lean.

- **Ankle and muscle timing improvements at lighter weights ($w = 0.1$):** On the more redundant H1622 and H2190 models, the clearest benefits shift to constraining the prior more. At $w = 0.1$, the prior rescues the ankle trajectory from baseline failures—achieving high ankle match fractions ($E_y = 0.57$ on H1622 and peaking at $E_y = 0.68$ on H2190) alongside positive correlations. Concurrently, it restores reciprocal activation timing in critical proximal and distal muscles such as the vasti (VL, VM) and hamstrings (BF, ST), with correlations reaching up to 0.88 with human reference patterns.

- **Muscle Effort:** Beyond spatial and temporal alignment, the prior alters total muscle activation despite the absence of an explicit effort term in the velocity reward. Prior-guided variants expend approximately 10–15% less total muscle activation across gait cycles.

- **Localization and structural boundaries:** These effects are not uniform across the musculature. Muscle groups lacking surface-EMG coverage are controlled primarily through the unconstrained residual and show no consistent

improvement across variants. Likewise, fiber lengths remain invariant across conditions, reflecting their stronger physical dependence on joint kinematics rather than the activation timing modified by the prior. The resulting picture is therefore not one of globally more human-like movement, but of localized changes strictly within the dimensions of coordination represented by the prior.

6.3 The Residual Weight as a Trade-off

The residual weight w governs the degree to which the human-derived coordination template dictates the final policy. Its impact varies non-monotonically across models and evaluation metrics. At $w = 0.5$, the prior preserves sufficient residual freedom for efficient learning and stable locomotion, while yielding optimal trajectory matching and kinetic push-off on H0918. However, this stronger constraint can override finer temporal features; on H2190, the localized ankle and hamstring advantages secured at $w = 0.1$ are diminished under the heavier 0.5 weighting.

Conversely, at $w \leq 0.1$, the prior exerts a dominant influence on internal muscle coordination, but exacts significant operational costs like slower convergence, elevated fall rates, and on H1622, a degenerate hopping gait across three of four random seeds of prior with $w = 0$. As demonstrated in Appendix 9.7, relaxing the constraint on a single muscle within a critical agonist–antagonist pair at $w = 0.1$ immediately eliminates this hopping behavior. This indicates that failure stems from holding specific individual muscles too rigidly to the decoder template rather than from an inherent flaw in the residual framework.

Henceforth, no single value of w optimizes all performance objectives simultaneously. While our findings align with Hausdörfer et al. (Hausdörfer et al., 2024) who advocate for larger residual weights to stabilize complex agents, constraining the prior with lower w also yields specific benefits.

6.4 Implications for Evaluating Musculoskeletal Control

These results demonstrate why task return alone provides an incomplete picture when evaluating learned musculoskeletal control. On H0918 for instance, all variants achieve maximum task return without falling despite exhibiting substantial differences in internal coordination, such as distinct ankle activation patterns, knee force profiles and — on H1622, qualitatively different locomotion strategies. This underscores that an under-determined task objective naturally admits multiple mechanically distinct solutions that achieve identical performance.

Consequently, a comprehensive assessment of musculoskeletal models requires

evaluating task performance alongside kinematics, kinetics, and internal muscle activity. The metrics used in this work provide complementary perspectives: the correlation coefficient r captures temporal agreement in shape and timing, whereas the match fraction E_y accounts for amplitude alignment. Combining these measures makes it possible to detect nuanced differences, such as a joint maintaining correct timing despite a shifted range of motion. Furthermore, evaluating muscle-level data requires accounting for the substantial inter-subject variability inherent in recordings.

Ultimately, these findings show that EMG-derived action priors offer an effective way to navigate this redundant solution space by biasing policies toward experimentally observed coordination patterns without relying on hand-crafted reward terms or low-level controllers.

6.5 Limitations and Future Work

Subjects and data. The prior is trained on the EMG of one healthy subject from the Camargo dataset, and the gaits are compared with six subjects of similar body size for kinematics and kinetics and with 22 subjects for muscle activity. Priors from more diverse populations, including other body sizes, ages, and pathological gait, and from other datasets are a natural next step.

Seeds. Each condition was trained with three, which is at the lower end for deep reinforcement learning, and several results vary strongly between seeds. More seeds per condition would allow tighter confidence intervals and significance tests.

Combining the prior with other sources of structure. The prior and DEP address the same problem, coordinating many muscles, from opposite directions: the prior takes the coordination from human EMG, while DEP builds it online from the agent's own muscle-length dynamics. The two could be combined, for example by letting DEP drive exploration in the residual or latent action while the decoder supplies the human pattern, so that the policy explores with correlated muscle activity from the start and is still guided towards human coordination. A second direction addresses the limited coverage of surface EMG. Synergies estimated by inverse simulation, as used by Park et al. (Park et al., 2026), cover every muscle of the model but are derived from kinematics rather than measured activity. A hybrid prior that uses EMG for the recorded muscles and inverse-simulation estimates for the deep and unrecorded ones, such as the hip flexors and adductors, would extend the prior to the whole leg, which matters most on H2190, where the EMG prior currently covers only a quarter of each leg.

Reward. We used a simple velocity reward on purpose to isolate the effect of the prior. The full reward, which guides the policy towards a natural gait, was tested only for the prior at $w = 0.1$ on H0918 and H1622. A broader and tuned comparison under such rewards would show whether the prior and gait-shaping objectives are complementary.

Temporal structure. The prior is only spatial, as the decoder maps single frames and specifies which muscles are active together, not when. Adding a temporal component, for example a gait-phase input to the decoder so that the prior follows the gait cycle, or noise models that capture temporal correlation, is a promising extension.

7. Conclusion

This thesis investigated whether surface-EMG-derived action priors shape muscle coordination in deep reinforcement learning approaches across musculoskeletal models with increasing complexity. To evaluate both task success and execution strategies, we assessed learned policies across multiple biomechanical metrics like joint kinematics, kinetics, and muscle activity.

The key result is that the prior does not primarily improve task learning itself. The state-of-the-art baselines are already capable of achieving the locomotion objective across the evaluated models under a well-defined velocity reward. Accordingly, the prior does not consistently improve task return or convergence speed. Its effect is instead visible in the solution selected by the policy. By constraining the action space according to coordination patterns extracted from sEMG, the prior is shown to bias the learned policy toward muscle activations that are more consistent with human locomotion.

This effect is most apparent in the kinematic and kinetic measures. With stronger prior weights ($w = 0.5$), the policy achieves improved trajectory matching at the hip and ankle, with match fractions of $E_y = 0.78$ and 0.61 on H0918, respectively, and produces a plantarflexor push-off moment with human-like magnitude and timing. With lighter weights ($w = 0.1$), the prior more strongly influences muscle activation timing, improving reciprocal coordination of muscles such as the vasti and hamstrings on the more redundant H1622 and H2190 models. Prior-guided variants also exhibit a 10–15% reduction in total muscle activation despite the absence of an explicit effort term in the task reward.

These results highlight the importance of balancing the prior with a learned residual. A sufficiently strong residual preserves the flexibility required for stable task learning, while a stronger influence of the prior allows the learned policy to more closely reflect the coordination encoded in the human data. However, overly restrictive constraints can introduce costs, including slower learning and increased falls during training, and can in some cases produce structurally undesirable gaits such as the hopping behavior observed on H1622. There is therefore no single prior strength that optimizes task performance and biomechanical similarity simultaneously.

Overall, these findings support using real human muscle data to bias the action space of reinforcement-learning policies in neuromusculoskeletal simulation. Rather than replacing task-driven learning, this prior shapes the otherwise under-

determined internal coordination of the learned solution. The results suggest that combining human-derived coordination with a learned residual recovers human-like features across multiple biomechanical measures while retaining the flexibility of reinforcement learning.

This provides a promising foundation for incorporating experimentally measured muscle coordination into computational models of human movement. Such an approach could be valuable in applications where physiological movement is important beyond task completion, including biomechanics research, rehabilitation, assistive-device design, and the study of neuromuscular control.

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9. Appendix

9.1 Implementation and Hardware

All agents are implemented in PyTorch on top of TonicRL (Pardo, 2020), using the MPO and DEP-MPO implementations and SCONE environments released with DEP-RL. The latent actio prior is implemented as an environment wrapper around these environments, so the learner itself is unchanged (Section 4.2). We use SCONE 2.4.1 with the Hyfydy 1.8.6 physics engine for simulations.

All policies were trained on a workstation with an AMD Ryzen Threadripper 3990X (64 cores), an NVIDIA RTX A6000 GPU and 125 GB of memory, with 200 environments in parallel per training. Training for 2×10^7 environment steps takes about 10 hours on H0918 and H1622 and about 28 hours on H2190.

9.2 EMG Processing

Step	Setting
Sampling rate	1000 Hz
High-pass filter	10 Hz, Butterworth, order 4, zero-phase
Band-pass filter	10–450 Hz, FIR, order 20, zero-phase
Rectification	absolute value
Envelope	low-pass 6 Hz, Butterworth, order 4, zero-phase
Segmentation	right heel strike to right heel strike, resampled to 101 points
Amplitude normalization	per subject and channel, min–max of mean treadmill stride (1.35 m/s)
Decoder input scaling	$\div$ mean per-stride 99th percentile, clipped to $[0, 1]$

Table 9.1: **EMG processing.** The first six steps follow the processing recommendations of the Camargo dataset.

The 11 channels and their abbreviations are gastrocnemius medialis (GAS), soleus (SOL), tibialis anterior (TA), vastus medialis (VM), vastus lateralis (VL), rectus femoris (RF), biceps femoris (BF), semitendinosus (ST), gracilis (GRA), gluteus medius (GMED) and external oblique (EO).

9.3 Evaluation Subjects

Dataset. This work uses the EMG of AB06 to train the decoder, the joint angles, moments and ground reaction forces of the six subjects of Table 9.2 as the kinematic

and kinetic reference, and the EMG of all 22 subjects as the reference for muscle activation timing.

Subject	Sex	Age	Height (m)	Mass (kg)	Treadmill	Level ground	Ramp	Stair
AB06*	M	20	1.80	74.8	961	303	468	252
AB08	M	21	1.74	72.6	811	364	431	210
AB10	M	22	1.75	83.9	851	322	431	269
AB11	M	21	1.75	77.1	841	330	505	264
AB23	M	20	1.80	76.8	824	326	453	252
AB30	M	31	1.77	77.0	771	311	462	258
All 22	13 M / 9 F	21.6 ± 3.6	1.71 ± 0.07	68.5 ± 11.3				

Table 9.2: **Subjects used in this work.** Number of strides per ambulation mode after segmentation. *Decoder training subject. The six listed subjects are the six male subjects closest to AB06 in height and mass and form the evaluation reference.

9.4 Kinematic and Kinetic Reference Data

The human reference for joint kinematics and kinetics (Section 4.9.5) is built from the same Camargo recordings as the EMG, using the six male subjects closest to AB06 in height and mass (AB06, AB08, AB10, AB11, AB23 and AB30; Appendix 9.3) during treadmill walking at 1.35 ± 0.25 m/s. Strides are segmented at the right heel strikes provided with the dataset, which are detected from motion capture. All signals are low-pass filtered before being resampled to 101 points per stride, and each quantity is taken from the source listed in Table 9.3.

Quantity	Source	Processing
Hip, knee, ankle angle	goniometers, 1000 Hz	low-pass 20 Hz, Butterworth order 4, zero-phase
Hip, knee, ankle moment	dataset inverse dynamics	÷ body mass; low-pass as above
Vertical GRF	force plate, right belt	÷ body weight; low-pass as above

Table 9.3: **Processing of the human kinematic and kinetic reference.**

The goniometers measure relative joint angles and carry a subject-specific offset. Before the match fraction of Eq. (4.17) is computed for the joint angles, the simulated mean trajectory is therefore shifted by the difference between its cycle mean and that of the human mean, so that the comparison concerns the shape and timing of the trajectory rather than its absolute zero. The sign conventions of the dataset (hip flexion, knee flexion and ankle dorsiflexion) match those of SCONE, so no sign changes are needed. The simulated signals are processed in the same way. Joint angles and the vertical ground reaction force are recorded at 100 Hz and low-pass filtered with the same filter; the ground reaction force is normalized

by body weight, and net joint moments are the sum of the moments of all muscles crossing the joint, divided by the model mass of 74.5 kg. Cycles are segmented at heel strike from the vertical ground reaction force (Section 4.9.3), and both legs are pooled.

The EMG reference for muscle activation timing (Section 4.9.4) uses all 22 subjects during treadmill walking at 1.2 ± 0.15 m/s, processed as in Appendix 9.2.

9.5 Reference Group for Muscle Activation Timing

The muscle activation timing of Section 4.9.4 was recomputed against the six reference subjects of the kinematic comparison instead of all 22, with the same runs, checkpoints and episodes. Over the 160 combinations of model, variant and muscle, the mean correlation changed by 0.003 on average and by at most 0.12, and the ordering of the variants within each muscle was preserved (Spearman correlation between the two orderings: median 0.99, minimum 0.89). The human range, the 5th percentile of the 15 pairwise correlations between the six subjects, was narrower than that of the 231 pairs of the full cohort. The within-range verdict therefore changed in 28 of the 160 combinations, 25 of them from inside to outside the range; half of the changes concerned TA. The changes were spread over all variants (MPO 5, DEP-MPO 5, EMG prior 11 over three values of w, untrained prior 3), so the choice of reference group does not favour any variant.

9.6 Hyperparameters

Hyperparameter	Value
Architecture	
Encoder / decoder layers	$11 \to 12 \to 6$ / $6 \to 12 \to 11$
Hidden / output activation	tanh / sigmoid
Latent dimension k	6
Loss	
Penalty threshold β / scale α	0.6 / 0.9
Penalty weight	1
Training	
Optimizer	Adam
Learning rate	10^{-3}
Batch size	512
Epochs	50
Training / held-out strides	1588 / 396 (AB06)
Training frames	50 000

Table 9.4: **Synergy decoder hyperparameters.**

Penalty and depth tuning. β, α and the depth were chosen on a grid over $\beta \in \{0.5, 0.6, \ldots, 1.0\}$, margin $\alpha - \beta \in \{0.1, 0.2, \ldots, 0.5\}$ and one or two hidden layers per side, with three seeds each, on the AB06 split. The 15 best settings reached held-out R^2 between 0.939 and 0.942. Averaged over the grid, one hidden layer reached 0.938 and two reached 0.935.

Single subject or pooled. All conditions used the architecture and penalty of Table 9.4, 50 000 training frames and the same 396 held-out AB06 strides, with the input scaling computed from each condition's own training data (Table 9.5).

Training data	Held-out AB06 R^2
AB06 alone	0.941 ± 0.001
All 22 subjects	0.898 ± 0.002
AB06 and the six subjects closest in height and mass	0.895 ± 0.003
21 subjects without AB06	0.890 ± 0.004

Table 9.5: **Single-subject and pooled decoders** (mean ± SD over three seeds). With AB08 as the target the ordering is the same (AB08 alone 0.904, pooled 0.876–0.887).

Policy and critic networks. Actor and critic use the same architecture for all variants and models (Table 9.6). Only the input and output sizes differ between

variants, since the policy output has dimension n for the baselines and $k + n$ for the prior variants.

Hyperparameter	Value
Actor	
Hidden layers	2×256, ReLU
Output distribution	Gaussian
Mean / SD activation	tanh / softplus
SD range	$[10^{-4}, 1]$
Critic	
Input	observation, action
Hidden layers	2×256, ReLU
Output	$Q(s, u)$
Input	
Observation normalization	running mean / SD

Table 9.6: **Actor and critic networks**, shared by all variants.

Training. All runs use the adaptive effort buffer, which subtracts $c \sum_i \rho_i^3$ from sampled rewards once the smoothed return exceeds 1000; under the velocity reward c therefore stays zero, while under the full reward it becomes active. DEP-MPO additionally uses the DEP settings of Table 9.8.

Hyperparameter	Value
Gain κ	1000
Time constant τ_{DEP}	40
Bias rate	0.002
Buffer size	200
Time distance / sensor delay	5 / 1
Regularization	32
s4avg	2
Normalization / Q-norm	independent / L2
Intervention length / probability	8 / 0.00371

Table 9.8: **DEP hyperparameters** for DEP-MPO

Hyperparameter	Value
Optimization	
Optimizer	Adam
Actor / critic learning rate	3×10^{-4}
Dual variable learning rate	10^{-2}
Discount γ	0.99
Target update (Polyak) τ	0.005
Gradient clipping	none
MPO	
Sampled actions per state K	20
E-step KL bound ϵ	0.1
M-step KL bound, mean / covariance	10^{-3} / 10^{-6}
Action penalization KL bound	10^{-3}
Initial log temperature	1
Initial log dual, mean / covariance	1 / 10
Minimum log dual	-18
Replay	
Buffer size	10^6
Batch size B	256
Return steps	1
Steps before first update	10^5 (2×10^5 DEP-MPO)
Env. steps / update iterations per cycle	1000 / 30
Schedule	
Parallel environments	200
Training budget	2×10^7 steps
Evaluation	every 2×10^5 steps, 20 episodes
Checkpoints	every 2×10^6 steps
Re-evaluation at ℓ_s^*	100 episodes

Table 9.7: **MPO and training hyperparameters**, shared by all variants.

Parameter	Value
Latent dimension k	6
Residual weight w on covered muscles	0, 0.1, 0.5 (EMG and untrained prior)
Residual weight on uncovered muscles	1
Residual a_{full}	raw policy output in $[-1, 1]$, not rescaled
Left leg	copies the right-leg rows of M
Policy output dimension $k + n$	24 (H0918), 28 (H1622), 96 (H2190)
Latent rescaling	$[-1, 1] \to [a_l^{\text{lo}}, a_l^{\text{hi}}]$ per dimension, Eq. (4.6)

Table 9.9: **Latent action prior.**

Control variants. The null prior's performance tracked very similar to like MPO, which shows that the wrapper and the additional latent outputs alone do not change learning so it is not reported further. The untrained decoder was initialized independently in each parallel training environment, so the policy did not learn with one fixed random decoder but with many. It therefore does not isolate the EMG content as intended, and its results are not reported beyond

Figure 5.1.

The null prior uses the same wrapper with $\hat{a}_t \equiv 0$ and residual weight 1 on every muscle, so the latent part of the policy output has no effect and $a_t = a_{\text{full},t}$ before clipping. The untrained prior uses a decoder with the architecture of Table 9.4 whose weights keep the default PyTorch initialization, drawn independently in each training environment, together with the latent range $[a_l^{\text{lo}}, a_l^{\text{hi}}]$ of the trained decoder, so that its latent actions are rescaled in the same way as for the EMG prior.

9.7 Relaxing the Prior on Agonist–Antagonist Muscles

At $w = 0.1$, the covered muscles can deviate from the decoded excitation by at most 0.1, and on H1622 three of four seeds hop (Section 5.6). To test whether a single over-constrained muscle causes this, we trained the prior on H1622 at $w = 0.1$ with one muscle per leg given a residual weight of 0.5 instead: the gluteus medius, which stabilizes the pelvis in the frontal plane, or the tibialis anterior, the antagonist of the plantarflexors. Two seeds were trained for each, and all were evaluated at the end of training (Table 9.10).

Variant (H1622)	Final return	Falls	Hops	E_y hip / knee / ankle			r GMED / TA
MPO	931	0%	no	0.36	0.30	0.12	0.41 / 0.60
EMG prior, $w = 0$	895	15%	3 of 4 seeds	0.33	0.31	0.30	0.79 / 0.32
EMG prior, $w = 0$, GMED at 0.5	913	20%	no	0.30	0.62	0.17	0.75 / 0.22
EMG prior, $w = 0$, TA at 0.5	930	0%	no	0.51	0.45	0.40	0.57 / 0.41

Table 9.10: **Loosening single muscles on H1622.** Final return: mean of the last ten training evaluations. Falls: fraction of recorded episodes ending in a fall. E_y and r as in Tables 5.4

Loosening either muscle removes the hopping gait: both variants walk, with a flight fraction below 0.1. Loosening TA gives the clearest improvement. It walks without falling, reaches the return of MPO, and matches all three joint angles better than MPO (hip E_y 0.51 against 0.36, ankle 0.40 against 0.12). Loosening GMED improves the knee (0.62) but falls in 20% of episodes. Both keep the human-like timing of GMED that the prior produces ($r = 0.57$–0.75, within the human range of 0.45). The hopping at $w = 0.1$ therefore does not follow from the prior as a whole, but from holding individual muscles too close to the decoded pattern, which the residual weight of a single muscle can relieve. These are two seeds per variant on one model, evaluated at the end of training, so we report them as an indication rather than a result.

9.8 Network Width

To test whether the larger models need more network capacity, MPO was also trained with actor and critic hidden layers of 2×512 instead of 2×256 units, one seed per model, with all other settings unchanged (Table 9.11).

Model	Width	Training return	Test return	Effort
H0918	256	942	990	0.42
	512	913	991	0.41
H1622	256	702	988	0.41
	512	626	893	0.39
H2190	256	499	412	0.40
	512	464	259	0.43

Table 9.11: **MPO with 2×256 and 2×512 hidden units** after 2×10^7 steps. Training return: mean stochastic training return; test return: deterministic evaluation return.

The wider network is not better on any model. On H1622 and H2190 its test return is lower (893 against 988 and 259 against 412), and its training return is lower on all three models. This points to slower optimization within the same step budget, not to missing capacity. All experiments therefore use 2×256 units.

9.9 Training Runs

Condition	H0918	H1622	H2190
MPO	3	3	3
DEP-MPO	3	3	2
EMG prior, $w = 0$	3	4	2
EMG prior, $w = 0.1$	3	6	4
EMG prior, $w = 0.5$	3	3	2
Untrained prior, $w = 0.5$	2	2	2
Null prior	3	3	2
Full reward: MPO / DEP-MPO / EMG prior $w = 0.1$	3 / 3 / 3	3 / 3 / 3	–
Sensitivity: $k \in \{2, 4, 8, 11\}$ at $w = 0.1$; $w \in \{0.2, 0.3\}$ at $k = 6$	3 each	–	–

Table 9.12: **Number of seeds per condition.** Velocity reward unless stated otherwise. All runs use a budget of 2×10^7 environment steps; checkpoints beyond it are not considered.

9.10 EMG to Muscle Mapping

Model	Model muscle (right leg)	EMG channel(s)
H0918	gastroc	GAS
	soleus	SOL
	tib_ant	TA
	vasti	VM, VL
	rect_fem	RF
	hamstrings	BF, ST
H1622	as H0918, plus glut_med	GMED
H2190	gas_med	GAS
	soleus	SOL
	tib_ant	TA
	vas_med / vas_lat	VM / VL
	rect_fem	RF
	bifemlh / semiten	BF / ST
	grac	GRA
	glut_med1	GMED
	ext_obl	EO

Table 9.13: **Mapping of EMG channels to model muscles.** A model muscle covered by several channels receives their average. All other muscles receive no prior. The left leg copies the right. Coverage per leg: H0918 6 of 9 muscles (67%), H1622 7 of 11 (64%), H2190 11 of 45 (24%).

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